

# Minkowski Bounds for Pasten’s Arithmetic Derivative Lattices in the Squarefree Subfamily

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## Abstract

For each coprime triple  $(a, b, c)$  with  $a + b = c$ , Pasten (2021) constructs a *universal derivative lattice*  $F(a, b)$  of rank  $\omega(abc) - 1$  from the linear constraint imposed by the weighted arithmetic derivative. We prove that for *squarefree* coprime triples, the lattice determinant satisfies  $\det(F(a, b)) < \text{rad}(abc)$ . Combined with Vaaler’s subspace lattice theorem, this yields a nonzero lattice vector with  $\ell^\infty$ -norm  $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$ .

We give a structural classification of partition types by whether the minimum *non-degenerate* norm ratio  $\rho = \|\psi_{\text{nd}}\|_\infty / R^{1/(\omega-1)}$  is universally bounded: nine types are bounded. We prove unconditional upper bounds, including  $2^{-1/(\omega-1)}$  for balanced types at  $\omega = 3, 4, 5$ ,  $\rho < 1$  for two  $\omega = 5$  types with  $a = 1$ ,  $\rho < 2^{-1/4}$  for type  $(2, 1, 2)$ , and  $\rho < 6^{-1/4}$  for type  $(2, 2, 1)$ . For  $\omega \geq 6$  we prove an explicit unboundedness criterion; applying it to every partition type requires the corresponding infinite prime-pattern family. The asymptotic constants are sharp only under the relevant prime-pattern conjectures. The inequalities have been checked over 44 000 squarefree triples and the key inequalities formalized in Lean 4 (zero sorry).

We prove a clean quality boundary: for squarefree triples with  $a \leq b$ , quality  $> 1/2$  if and only if  $a = 1$ . Combined with the classification, this shows that the triples hardest for abc (highest quality) are structurally opposite to the triples closest to the extremal  $\rho$  bound in the surveyed families. The argument is self-contained and uses no abc-equivalent hypothesis.

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**Honest scope.** The result proved here is a structural property of Pasten’s lattice. It does *not* prove the abc conjecture, nor does it provide progress toward abc for the hard cases (high-quality triples, which have small  $\omega$ ). The abc conjecture remains [OUT] in the sense of the accompanying verification kernel.

# 1 Introduction

## 1.1 Background

The abc conjecture of Masser–Oesterlé asserts that for every  $\varepsilon > 0$  there exists  $K_\varepsilon > 0$  such that for all coprime positive integers  $a, b, c$  with  $a + b = c$ :

$$c \leq K_\varepsilon \cdot \text{rad}(abc)^{1+\varepsilon},$$

where  $\text{rad}(n) = \prod_{p|n} p$  is the radical. This remains an open problem.

Pasten [1] introduced a framework relating the abc conjecture to the geometry of numbers via the *universal arithmetic derivative*. For each prime  $p$  and weight  $\psi(p) \in \mathbb{Z}$ , the  $\psi$ -arithmetic derivative is

$$d^\psi(n) = n \sum_{p|n} \frac{v_p(n)}{p} \psi(p),$$

where  $v_p(n)$  is the  $p$ -adic valuation. The key object is the *lattice of additive weights*:

$$F(a, b) = \{ \psi : \mathbb{P}(abc) \rightarrow \mathbb{Z} \mid d^\psi(a) + d^\psi(b) = d^\psi(c) \}.$$

Since  $\gcd(a, b) = 1$  and  $a + b = c$ , the prime supports  $P_a, P_b, P_c$  of  $a, b, c$  are pairwise disjoint (for squarefree triples), and the additivity constraint reduces to a single integer linear equation. Pasten showed that  $F(a, b)$  has rank  $\omega(abc) - 1$ , where  $\omega(n)$  denotes the number of distinct prime divisors of  $n$ .

Pasten's *Small Derivatives Conjecture* (SDC) asserts that, apart from finitely many triples and from triples of the form  $(1, N, q)$  with  $q$  prime (up to order), the minimum  $\ell^\infty$ -norm of a non-degenerate vector in  $F(a, b)$  is at most  $c^\eta$  for some  $\eta < 1$ . Pasten proved that SDC and the abc conjecture are equivalent with an explicit relation of exponents. Siegel's lemma gives only  $\eta = 1$  (the trivial bound); beating this would be a breakthrough.

Our contribution is more modest: we establish the exact determinant of  $F(a, b)$  for squarefree triples, and derive the corresponding Minkowski bound  $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$ . This is the  $\eta = 1/(\omega-1)$  *bound relative to*  $\text{rad}(abc)$ , which is non-trivial when  $\omega \geq 3$ . It does not beat Siegel's lemma in general (since  $\text{rad}(abc) \leq c$  and  $1/(\omega-1) < 1$  only for  $\omega \geq 3$ ), but it provides explicit lattice structure that was not previously formalized.

## 1.2 Main results

Let  $(a, b, c)$  be a coprime squarefree triple with  $a + b = c$  and  $\omega = \omega(abc) \geq 2$ . Set  $P = P_a \cup P_b \cup P_c$  and  $R = \text{rad}(abc) = \prod_{p \in P} p$ .

**Theorem 1** (Determinant bound).  $\det(F(a, b)) = R \cdot \sqrt{\sum_{p \in P} p^{-2}} < R$ .

**Theorem 2** (GCD lemma). *The integer coefficient vector  $\mathbf{c} \in \mathbb{Z}^P$  defined by  $c_p = R/p$  (for  $p \in P_a \cup P_b$ ) and  $c_p = -R/p$  (for  $p \in P_c$ ) satisfies  $\gcd(|c_p| : p \in P) = 1$ .*

**Corollary 3** (Vaaler lattice bound [2]). *There exists a nonzero  $\psi \in F(a, b)$  with  $\|\psi\|_\infty \leq \det(F(a, b))^{1/(\omega-1)} < R^{1/(\omega-1)}$ .*

For non-squarefree triples, Theorem 15 below gives the upper bound  $\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}$  for Pasten's original lattice, where  $v_{\max} = \max_p v_p(abc)$ . The finer exact formulas introduced for non-squarefree triples apply to a valuation-coordinate sublattice defined below, not to the original lattice without further argument.

Corollary 3 does *not* imply abc for the following reasons:

1. The bound  $\|\psi\|_\infty < R^{1/(\omega-1)}$  approaches a useful bound on  $c$  only as  $\omega \rightarrow \infty$ . The hard cases for abc are triples with *small*  $\omega$  (prime powers, Mersenne-type), where  $R^{1/(\omega-1)}$  is close to  $R \leq c$ —the trivial bound.
2. The vector  $\psi$  produced by Corollary 3 bounds the *lattice* minimum, not the height of  $c$  directly. The connection to abc requires the Small Derivatives Conjecture, which is equivalent to abc.
3. For the degenerate  $\omega = 2$  family (e.g.  $c = 2^k$ ,  $a = 1$ ), the bound gives  $\|\psi\|_\infty \leq R^1 = R$ , which is the trivial bound. Pasten explicitly excludes this family from SDC for this reason.

## 2 Definitions and setup

### 2.1 The Pasten lattice for squarefree triples

Let  $(a, b, c)$  be a coprime squarefree triple,  $a + b = c$ ,  $a, b, c > 0$ . Write  $v_p(n) = 1$  for all  $p \mid n$  (squarefree hypothesis).

Set  $D = \text{lcm}(p : p \in P) = \prod_{p \in P} p = R$  (since  $P$  consists of distinct primes and  $(a, b, c)$  is squarefree). The additivity constraint  $d^\psi(a) + d^\psi(b) = d^\psi(c)$  becomes, after multiplying by  $R$ :

$$\sum_{p \in P} c_p \cdot \psi(p) = 0,$$

where

$$c_p = \begin{cases} +R/p & p \in P_a \cup P_b, \\ -R/p & p \in P_c. \end{cases}$$

Note  $c_p \in \mathbb{Z}$  since  $p \mid R$ . The lattice is

$$F(a, b) = L = \{\psi \in \mathbb{Z}^P \mid \mathbf{c} \cdot \psi = 0\},$$

a rank- $(\omega - 1)$  sublattice of  $\mathbb{Z}^P$ .

## 2.2 Lattice determinant

For an integer lattice defined by a single primitive constraint  $\mathbf{c} \in \mathbb{Z}^\omega$  with  $\text{gcd}(c_p) = 1$ , the determinant equals

$$\det(L) = \|\mathbf{c}\|_2 = \left( \sum_{p \in P} c_p^2 \right)^{1/2}.$$

If  $\text{gcd}(c_p) = g > 1$ , the primitive form is  $\mathbf{c}/g$  and  $\det(L) = \|\mathbf{c}\|_2/g$ .

## 3 Proofs

### 3.1 Proof of Theorem 2

*Proof.* Each  $|c_p| = R/p = \prod_{q \in P, q \neq p} q$ . Suppose a prime  $\ell$  divides  $\text{gcd}(|c_p| : p \in P)$ . Since  $\ell$  divides  $|c_p|$  for every  $p \in P$ , in particular  $\ell$  divides  $|c_p|$  for some  $p$ ; since  $|c_p| = R/p$  is a product of primes from  $P$ , we must have  $\ell \in P$ .

Take  $p = \ell$ : then  $|c_\ell| = R/\ell = \prod_{q \in P, q \neq \ell} q$ . This product does *not* contain  $\ell$  as a factor, so  $\ell \nmid R/\ell$ . But  $\ell$  was assumed to divide every  $|c_p|$ , including  $|c_\ell|$ . Contradiction. Hence  $\text{gcd}(|c_p|) = 1$ .  $\square$

**Remark 1.** *Edge case*  $|P| = 2$ ,  $P = \{p, q\}$ :  $|c_p| = q$ ,  $|c_q| = p$ ,  $\text{gcd}(p, q) = 1$ . *The case*  $|P| = 1$  *is excluded by*  $\omega \geq 2$ .

### 3.2 Proof of Theorem 1

*Proof.* By Theorem 2,  $\text{gcd}(|c_p| : p \in P) = 1$ , so  $\det(L) = \|\mathbf{c}\|_2$ . Since  $c_p = \pm R/p$  we have  $c_p^2 = (R/p)^2 = R^2/p^2$ , thus

$$\|\mathbf{c}\|_2^2 = \sum_{p \in P} \frac{R^2}{p^2} = R^2 \sum_{p \in P} \frac{1}{p^2}.$$

So  $\det(L) = R \sqrt{\sum_{p \in P} p^{-2}}$ .

It remains to show  $\sum_{p \in P} p^{-2} < 1$ . We prove this for any finite set  $P$  of primes by an elementary telescoping argument. Set  $M = \max P$ . For each  $n \geq 2$ :

$$\frac{1}{n^2} \leq \frac{1}{n-1} - \frac{1}{n},$$

which follows from the identity  $\frac{1}{n-1} - \frac{1}{n} - \frac{1}{n^2} = \frac{1}{n^2(n-1)} > 0$ . Therefore

$$\sum_{p \in P} \frac{1}{p^2} \leq \sum_{p \in P} \left( \frac{1}{p-1} - \frac{1}{p} \right) \leq \sum_{n=2}^M \left( \frac{1}{n-1} - \frac{1}{n} \right) = 1 - \frac{1}{M} < 1.$$

(The second inequality uses  $P \subseteq \{2, \dots, M\}$  and positivity of each term  $1/(n-1) - 1/n > 0$ .) Hence  $\det(L) < R$ .  $\square$

**Remark 2.** The true value is  $\sum_{p \text{ prime}} p^{-2} = P(2) \approx 0.4522$  (prime zeta function at  $s = 2$ ). An elementary bound gives  $\sum_{p \text{ prime}} p^{-2} \leq 11/18 \approx 0.611 < 1$  via  $1/4 + 1/9 + \int_4^\infty x^{-2} dx = 1/4 + 1/9 + 1/4 = 11/18$ .

### 3.3 Proof of Corollary 3

*Proof.* By Theorem 1,  $\det(L) < R$ . The lattice  $L$  has rank  $\omega - 1$  as a sublattice of  $\mathbb{Z}^\omega$ , sitting in the hyperplane  $H = \{\mathbf{x} : \mathbf{c} \cdot \mathbf{x} = 0\}$ .

We use Theorem 2 of Vaaler [2], which gives a precise form of Minkowski's linear-forms theorem in the subspace setting. Let  $A$  be the  $\omega \times (\omega - 1)$  matrix whose columns form a  $\mathbb{Z}$ -basis of  $L$ , and take the  $\omega$  ambient coordinate functionals as the linear forms. Then  $\det(A^T A) = \det(L)^2$ . Vaaler's theorem, with all coordinate bounds equal to  $\lambda := \det(L)^{1/(\omega-1)}$ , gives a nonzero  $\psi \in L$  with  $\|\psi\|_\infty \leq \lambda$ . Equivalently, its geometric corollary says that every central hyperplane section of  $[-1, 1]^\omega$  has  $(\omega - 1)$ -dimensional volume at least  $2^{\omega-1}$ . Therefore

$$\|\psi\|_\infty \leq \det(L)^{1/(\omega-1)} < R^{1/(\omega-1)}.$$

$\square$

## 4 Numerical evidence

Table 1 gives comprehensive statistics from script `t30_rho_distribution.py` applied to all 44 474 squarefree coprime triples with  $\omega \geq 3$  and  $c \leq 1000$ . Table 2 breaks out the nine universally-bounded partition types.

The unbounded types drive the high mean  $\rho$  for  $\omega \leq 5$ . For the bounded partition types, the distribution concentrates well below the proved upper bounds: the median  $\rho$  for combined  $\omega = 4$  bounded types is 0.34, and fewer than 1% of triples come within 0.02 of  $2^{-1/3}$ .

| $\omega$ | count  | mean $\rho$ | max $\rho$ | % > 1 | note                                      |
|----------|--------|-------------|------------|-------|-------------------------------------------|
| 3        | 79     | 0.864       | 1.080      | 30.4% | includes unbounded types (0, 2, 1), etc.  |
| 4        | 3 223  | 0.945       | 9.919      | 27.7% | (2, 1, 1), (0, 1, 3), (0, 3, 1) unbounded |
| 5        | 14 009 | 0.961       | 31.105     | 24.1% | (1, 1, 3), (1, 3, 1), (3, 1, 1) unbounded |
| 6        | 18 739 | 0.465       | 17.889     | 5.3%  | all types unbounded                       |
| 7        | 7 766  | 0.248       | 1.631      | 1.0%  | all types unbounded                       |

Table 1: Empirical summary of  $\rho = \|\psi_{\text{nd}}\|_{\infty}/R^{1/(\omega-1)}$  organized by  $\omega = \omega(abc)$  over all squarefree coprime triples with  $c \leq 1000$ . High mean  $\rho$  for  $\omega = 3, 4, 5$  is driven by unbounded partition types.

| Type                    | $n$   | mean $\rho$ | max $\rho$ | upper bound                        | % > 0.5 | status  |
|-------------------------|-------|-------------|------------|------------------------------------|---------|---------|
| (1, 1, 1), $\omega = 3$ | 35    | 0.693       | 0.706      | $2^{-1/2} \approx 0.707$           | 100%    | BOUNDED |
| (0, 2, 2), $\omega = 4$ | 69    | 0.147       | 0.505      | $3 \cdot 210^{-1/3} \approx 0.505$ | 1.4%    | BOUNDED |
| (1, 1, 2), $\omega = 4$ | 1 029 | 0.356       | 0.781      | $2^{-1/3} \approx 0.794$           | 30.0%   | BOUNDED |
| (1, 2, 1), $\omega = 4$ | 1 036 | 0.372       | 0.791      | $2^{-1/3} \approx 0.794$           | 31.9%   | BOUNDED |
| (1, 2, 2), $\omega = 5$ | 3 450 | 0.104       | 0.817      | $2^{-1/4} \approx 0.8408$          | 0.0%    | BOUNDED |
| (2, 1, 2), $\omega = 5$ | 3 115 | 0.093       | 0.634      | $2^{-1/4} \approx 0.841$           | 0.3%    | BOUNDED |
| (2, 2, 1), $\omega = 5$ | 3 006 | 0.089       | 0.611      | $6^{-1/4} \approx 0.6389$          | 0.4%    | BOUNDED |
| (0, 2, 3), $\omega = 5$ | —     | —           | —          | 1                                  | —       | BOUNDED |
| (0, 3, 2), $\omega = 5$ | —     | —           | —          | 1                                  | —       | BOUNDED |

Table 2: Distribution of  $\rho$  for the nine universally-bounded partition types,  $c \leq 1000$  (for  $\omega \leq 4$  and  $\omega = 5$  multi-group types). The displayed constants are proved upper bounds. Equality in an asymptotic bound would require an infinite prime-pattern family (such as twin primes); sharpness is therefore conditional and is not asserted by the table. P99 and P100 for combined  $\omega = 4$  bounded types: 0.775 and 0.791, both below  $2^{-1/3} = 0.794$ .

## 5 Discussion

### 5.1 The degenerate $\omega = 2$ family

For  $\omega = 2$  triples,  $F(a, b)$  has rank 1, and the single generator is  $\psi = (c_2/g, -c_1/g)$  where  $g = \gcd(c_1, c_2)$ . In the squarefree case  $P = \{p, q\}$  and the minimum norm is  $\max(q, p) = \max(p, q)$ . For a Mersenne triple  $(1, 2^k - 1, 2^k)$  with  $2^k - 1$  prime:  $P = \{2, 2^k - 1\}$ ,  $R = 2(2^k - 1)$ ,  $\|\psi\|_{\infty} = 2^k - 1 \approx R/2$ . So  $\|\psi\|_{\infty}/R \rightarrow 1/2$ , not smaller. The bound is essentially tight.

### 5.2 Why this does not prove abc

Let  $(a, b, c)$  be a high-quality abc triple (quality =  $\log c / \log R > 1$ ). High-quality triples tend to have small  $\omega$  (often  $\omega = 3$  or 4). Corollary 3 gives  $\|\psi\|_{\infty} \leq R^{1/(\omega-1)}$ . For  $\omega = 3$ :  $\|\psi\|_{\infty} \leq R^{1/2}$ . Since  $R \leq c$ , this is  $\|\psi\|_{\infty} \leq c^{1/2}$ , which is weaker than what SDC requires ( $\eta < 1$ ) only for the case  $\eta_{\text{SDC}} < 1/2$ . The abc conjecture is equivalent to SDC holding for all  $\varepsilon > 0$ , i.e.  $\|\psi\|_{\infty} = o(c^{\varepsilon})$  for all  $\varepsilon > 0$  — an exponential improvement over what we prove.

The gap between our Minkowski bound ( $\eta_R = 1/(\omega - 1)$ ) and SDC ( $\eta_c = \varepsilon$  for all  $\varepsilon > 0$ )

is the full depth of  $abc$ .

### 5.3 Quality boundary for squarefree triples

For squarefree coprime  $(a, b, c)$  with  $a + b = c$  and  $R = \text{rad}(abc) = abc$ , the quality satisfies  $\text{quality} = \log c / \log(abc)$ . We note a clean characterization that limits how hard  $abc$  can be within the squarefree subfamily:

**Proposition 4** (Quality boundary). *For squarefree coprime  $(a, b, c)$  with  $a + b = c$  and  $a \leq b$ :*

$$\text{quality} > \frac{1}{2} \iff a = 1.$$

*Moreover,  $\text{quality} < 1$  always, and  $\text{quality} = \frac{1}{2}$  is never achieved. For  $a = 1$ ,  $\text{quality} \rightarrow \frac{1}{2}$  from above as  $b \rightarrow \infty$ . For  $a = 2$ ,  $\text{quality} \rightarrow \frac{1}{2}$  from below.*

*Proof.*  $\text{quality} > \frac{1}{2}$  iff  $\log c > \log a + \log b$ , i.e.  $c > ab$ , i.e.  $a + b > ab$ , i.e.  $(a - 1)(b - 1) < 1$ . For positive integers, this means  $(a - 1)(b - 1) = 0$ , i.e.  $a = 1$  or  $b = 1$ . Since  $a \leq b$  and  $a \geq 1$ :  $b = 1 \Rightarrow a = 1$ . So  $\text{quality} > 1/2$  iff  $a = 1$ .  $\text{Quality} < 1$  since  $R = abc > c$ .  $\text{Quality} = 1/2$  would require  $c = ab$ , impossible since  $c = a + b < ab$  for  $a, b \geq 2$  and  $c = a + b > ab = b$  for  $a = 1$ .  $\square$

The Lean 4 formalization of the combinatorial core ( $ab < a + b \leftrightarrow a = 1$  for  $1 \leq a \leq b$ ) is given by `pasten.F21A.ab_lt_sum_iff_a.one` (proved by `omega/nlinarith`, zero sorry).

### 5.4 Quality- $\rho$ trade-off

Proposition 4 implies that squarefree triples with  $\text{quality} > 1/2$  are exactly the *trivial* triples ( $a = 1$ ). For the harder non-trivial cases ( $a \geq 2$ ),  $\text{quality} \leq 1/2$  with equality only in the limit.

Within the sampled bounded partition types, quality and  $\rho$  point in *opposite* directions:

- **High- $\rho$  families** (balanced types approaching the conditional extremal bound): e.g. type  $(1, 1, 2)$  with  $a = p \approx q$  both large odd primes. Here  $\rho^3 \rightarrow 1/2$  (approaching  $2^{-1/3}$ ) while  $\text{quality} \rightarrow \log(2p) / \log(2p^3) = 1/3$ .
- **High-quality families** ( $a = 2$  fixed,  $b \rightarrow \infty$ ): Here  $\text{quality} \rightarrow 1/2$  (approaching the squarefree non-trivial supremum) while  $\rho = 2 / (2 \cdot b \cdot (2 + b))^{1/3} \rightarrow 0$ .

Empirically (Table 2,  $c \leq 2000$ ), Pearson  $r(\text{quality}, \rho) \approx -0.76$  for types  $(1, 1, 2)$  and  $(1, 2, 1)$ .

**Conclusion.** The triples hardest for  $abc$  (highest quality, approaching  $\text{quality} = 1/2$  from below) are not the triples closest to the extremal  $\rho$  bound in the surveyed families. The gap between “hard for  $abc$ ” and “hard for SDC” is a provable structural feature: within the squarefree subfamily, maximizing quality and maximizing  $\rho$  are *geometrically opposite* in the parameter space of partition types.

### F31: joint quality- $\rho$ bound

The opposition between quality and  $\rho$  is not merely empirical: for every *bounded- $\rho$*  partition type (those with  $\sup_{\text{triples}} \rho < \infty$ ) the sum  $\text{quality} + \rho^{\omega-1}$  is less than 1.

**Theorem 5** (F31: Quality- $\rho$  Joint Bound). *For every squarefree coprime triple  $(a, b, c)$  with  $a + b = c$ ,  $a \geq 2$ , belonging to a bounded- $\rho$  partition type:*

$$\text{quality}(a, b, c) + \rho(a, b, c)^{\omega-1} < 1.$$

*The bound is strict.*

*Proof.* By Proposition 4 (quality boundary),  $\text{quality} < 1/2$  for all non-trivial squarefree triples ( $a \geq 2$ ). For a bounded type containing a one-prime constituent, let  $g \geq 2$  be the smallest group minimum and  $d$  the second-smallest. Every remaining prime is larger than  $d$ , so  $R > 2d^{\omega-1}$  and  $\rho^{\omega-1} < 1/2$ . The only bounded type with  $a \geq 2$  and no one-prime constituent is  $(2, 2, 1)$ . In that case, writing  $g, h$  for the primes in the first constituent,  $d, q_2$  for those in the second, and  $r$  for the single third-prime, with  $d$  the second-smallest group minimum, gives  $ghq_2r > 6d^2$  and hence  $\rho^{\omega-1} = d^3/(ghq_2r) < 1/6$ . Hence  $\text{quality} + \rho^{\omega-1} < 1/2 + 1/2 = 1$ .  $\square$

**Remark 3** (Conditional tightness for type  $(1, 1, 1)$ ). *The bound would be tight along type  $(1, 1, 1)$  triples if infinitely many twin primes existed. For the twin-prime family  $(2, q, q+2)$ ,  $\text{quality} \rightarrow 1/2$  and  $\rho^2 = q/(2(q+2)) \rightarrow 1/2$  as  $q \rightarrow \infty$ . The infinitude of twin primes is open, so this is conditional sharpness rather than an unconditional determination of the supremum. The algebraic core —  $(q+2)^q < (2q)^{q+4}$  for all  $q \geq 1$  (proved for  $q = 1$  directly; for  $q \geq 2$  by  $q+2 \leq 2q$  and  $q < q+4$ ) — is formalized as `pasten_F31a` in the Lean kernel.*

**Remark 4** (Unbounded types). *The bound fails for unbounded- $\rho$  types (e.g. type  $(2, 1, 1)$ :  $\rho^3 \approx q/6 \rightarrow \infty$ ). For bounded types the gap to 1 is large away from type  $(1, 1, 1)$ :  $\omega = 4$  bounded types  $\max(\text{quality} + \rho^{\omega-1}) \approx 0.523$ ;  $\omega = 5$  bounded types  $\max(\text{quality} + \rho^{\omega-1}) \approx 0.569$  (verified numerically, T51). Under the F11 prime-pattern criterion,  $\rho^{\omega-1} \rightarrow \infty$  and the bound  $\text{quality} + \rho^{\omega-1} \leq 1$  fails along that sequence.*

## 5.5 Non-degeneracy: complete classification for $\omega \leq 5$

Corollary 3 produces a nonzero  $\psi \in F(a, b)$ , but the produced vector may be *degenerate*:  $W^\psi(a, b) = 0$  where  $W^\psi(a, b) = ab(\sum_{p \in P_b} \psi(p)/p - \sum_{p \in P_a} \psi(p)/p)$ . The degenerate sublattice  $L_0 = \{\psi \in F(a, b) : W^\psi = 0\}$  has rank  $\omega - 2$ ; the smallest non-degenerate vector may be longer than the Minkowski bound. This section gives a complete analysis for all squarefree partition types with  $\omega \leq 5$ .

### Universal Divisibility Lemma and coordinate change

**Theorem 6** (F8: Universal Divisibility). *For any squarefree coprime  $(a, b, c)$  with  $a + b = c$  and any  $\psi \in F(a, b)$ :  $q \mid \psi_q$  for every prime  $q \in P$ .*



*Proof.* The integer lattice constraint isolates  $\psi_q$ :  $(R/q)\psi_q = -\sum_{p \neq q} \text{sign}(p)(R/p)\psi_p$ . For each  $p \neq q$ :  $R/p$  contains  $q$  as a factor (since  $q \in P$ ,  $q \neq p$ ). Hence  $q \mid \text{RHS}$ . Since  $\gcd(q, R/q) = 1$ :  $q \mid \psi_q$ .  $\square$

Define  $\varphi_q = \psi_q/q \in \mathbb{Z}$  (well-defined by Theorem 6). This gives a canonical bijection

$$F(a, b) \cong \tilde{F}(a, b) = \{\varphi \in \mathbb{Z}^\omega \mid \sum_{q \in P} \text{sign}(q) \varphi_q = 0\},$$

where  $\text{sign}(q) = +1$  if  $q \mid ab$  and  $-1$  if  $q \mid c$ . The Wronskian becomes  $W = ab(S_b - S_a)$  where  $S_X = \sum_{q \mid X} \varphi_q$ . Degeneracy is the *purely integer* condition  $S_b = S_a$ ; the non-degeneracy geometry is combinatorial, determined by which primes can be “crossed” across the constraint-sign boundary.

### $\omega = 3$ : all non-degenerate

For squarefree prime triples  $(p, q, r)$  with  $p + q = r$  (parity forces  $p = 2$ ), the degenerate sublattice  $L_0$  is generated by  $(p, q, 2r)$  with  $\ell^\infty$ -norm  $2r$ . Combining Theorem 1 with the Minkowski–Vaaler bound gives a vector of  $\ell^\infty$ -norm at most  $\sqrt{R} = \sqrt{pqr} < 2r$  (proved:  $pq < 4r$  for prime triples, hence  $pqr < 4r^2$ ).

**Theorem 7** (F3:  $\omega=3$  non-degeneracy). *For every squarefree prime triple  $(p, q, r)$  with  $p + q = r$ , the shortest non-zero vector of  $F(p, q)$  is non-degenerate; equivalently,  $\text{nd}(p, q) < 2r$ .*

Therefore Theorem 7 says that the *shortest vector is always non-degenerate* for  $\omega = 3$  prime triples. Four constituent lemmas are formalized in Lean 4 (zero sorry).

### $\omega = 4$ : complete partition-type classification

For squarefree coprime  $\omega = 4$  triples with partition type  $(n_a, n_b, n_c)$ ,  $n_a + n_b + n_c = 4$ , all six types are classified analytically via the  $\varphi$ -coordinate framework (scripts **t14–t19**, verified for all triples  $c \leq 300$ ):

| Type      | Degen?      | Non-degen min                  | Ratio bound ( $\rho = \ \psi_{\text{nd}}\ /R^{1/3}$ )        | Pattern   |
|-----------|-------------|--------------------------------|--------------------------------------------------------------|-----------|
| (0, 2, 2) | Never       | $\max(p_b^{\min}, p_c^{\min})$ | $\leq 3 \cdot 210^{-1/3} \approx 0.505$ , max at (1, 14, 15) | BOUNDED   |
| (1, 1, 2) | Never       | $\max(p_a, p_b)$               | $< 2^{-1/3} \approx 0.794$ (F14)                             | BOUNDED   |
| (1, 2, 1) | iff $a > r$ | $\max(a, q)$                   | $< 2^{-1/3} \approx 0.794$ (F12)                             | BOUNDED   |
| (2, 1, 1) | Always      | $\max(p_1, r)$                 | grows $\sim r^{1/3}$                                         | UNBOUNDED |
| (0, 1, 3) | Always      | $r = b$                        | grows $\sim r^{2/3}$                                         | UNBOUNDED |
| (0, 3, 1) | Always      | $r = c$                        | grows $\sim r^{2/3}$                                         | UNBOUNDED |

Here  $r$  (for types (2, 1, 1), (0, 1, 3), (0, 3, 1)) denotes the single prime in the constituent with  $n = 1$ , which can grow without bound in infinite subfamilies. For type (1, 2, 1), the degeneracy condition  $a > r$  (where  $r$  is the larger prime of  $b$ ) forces  $r < a \leq 2r$ , and the ratio  $\max(a, q)/R^{1/3}$  is bounded by  $2^{-1/3}$  (proved by elementary calculus: the function  $t^{2/3}/(2(t+2))^{1/3}$  is strictly increasing on (1, 2) and remains below the conditional endpoint  $2^{-1/3}$  at  $t \rightarrow 2$ ).

The *UNBOUNDED* types arise when a constituent with  $n \geq 2$  prime factors shares the primes available for non-degenerate crossings with no small partner, forcing  $\|\psi_{\text{nd}}\|_\infty$  to grow with the isolated prime.

### $\omega = 5$ : all partition types

The  $\varphi$ -coordinate classification extends directly to  $\omega = 5$  (script `t20`, 902 triples  $c \leq 200$ ):

| Type      | Degen/ND | Max $\rho$ ( $= \ \psi_{\text{nd}}\ /R^{1/4}$ ) | Achieved at | Pattern         |
|-----------|----------|-------------------------------------------------|-------------|-----------------|
| (1, 2, 2) | 0/219    | $2^{-1/4} \approx 0.8408$ (upper bound)         | —           | BOUNDED (F26)   |
| (2, 1, 2) | 17/171   | $2^{-1/4} \approx 0.841$ (upper bound)          | —           | BOUNDED (F30)   |
| (2, 2, 1) | 15/189   | $6^{-1/4} \approx 0.6389$ (upper bound)         | —           | BOUNDED (F24)   |
| (0, 2, 3) | —        | $< 1$                                           | —           | BOUNDED (F23)   |
| (0, 3, 2) | —        | $< 1$                                           | —           | BOUNDED (F23)   |
| (1, 1, 3) | 112/118  | 2.798                                           | —           | UNBOUNDED (F11) |
| (1, 3, 1) | 106/110  | 2.798                                           | —           | UNBOUNDED (F11) |
| (3, 1, 1) | 0/123    | 6.745                                           | —           | UNBOUNDED (F11) |
| (0, 1, 4) | —        | $\rightarrow \infty$                            | —           | UNBOUNDED (F22) |
| (0, 4, 1) | —        | $\rightarrow \infty$                            | —           | UNBOUNDED (F22) |

All five bounded  $\omega = 5$  types have strict upper bounds. Their extremal constants are conditional on infinite prime-pattern families and are not asserted as unconditional suprema. Type (1, 2, 2) extends the universal bound  $2^{-1/(\omega-1)}$  to  $\omega = 5$ : for  $a = 2$ ,  $b = pq$ ,  $c = rs$  with all four odd primes near equal and  $2 + pq = rs$ ,  $\rho^4 = p^3/(2qrs) < 1/2$  since  $q, r, s > p$  (Theorem 8). As all four primes  $\rightarrow n$  in such a family:  $\rho^4 \rightarrow 1/2$ , giving conditional sharpness of  $2^{-1/4}$ . Verified: 0 violations of  $\rho < 2^{-1/4}$  for  $a = 2$ ,  $b \leq 200\,000$ . Types (2, 1, 2) and (1, 2, 2) share the conditional extremal constant  $2^{-1/4} \approx 0.8409$ : by the  $a \leftrightarrow b$  symmetry of  $\rho$  (which depends only on  $\{\min P_a, \min P_b, \min P_c\}$  and  $R$ ), any (2, 1, 2) triple  $(a, b, c)$  gives a (1, 2, 2) triple  $(b, a, c)$  with the same  $\rho$ . Hence the conditional constant for (2, 1, 2) equals that for (1, 2, 2). The extremal family:  $b = 2$  (fixed),  $a = p_1 p_2$ ,  $c = p_1 p_2 + 2 = r_1 r_2$  near-twin semiprimes;  $\text{nd} = p_1$ ,  $\rho^4 = p_1^3/(2 p_2 r_1 r_2) < 1/2$  (since  $p_2 r_1 r_2 \geq p_1 p_2^2 > p_1^3$ ). As  $p_1 \approx p_2 \approx r_1 \approx r_2 \rightarrow n$  in such a family:  $\rho^4 \rightarrow 1/2$ . Verified: 0 violations of  $\rho < 2^{-1/4}$  for 2,475,742 triples at  $c \leq 50,000$  (6 violations of the old incorrect bound  $6^{-1/4}$ ; the old F25 claim was wrong).

**Theorem 8** (F26: type (1, 2, 2) upper bound). *For any type (1, 2, 2) triple,  $\rho^4 < 1/2$ , so  $\rho < 2^{-1/4}$ . In the subfamily  $a = 2$ ,  $b = pq$ ,  $c = rs$  (with  $2 + pq = rs$ ,  $p \leq q$ , and  $r \leq s$ ),  $\rho^4 = p^3/(2qrs)$ . Equality is never attained. The constant  $2^{-1/4}$  would be sharp if the corresponding near-equal prime-pattern family were infinite; this sharpness is conditional and is not claimed as a theorem.*

*Proof.* Let  $g \geq 2$  be the single-prime group's minimum and let  $d$  be the second smallest of the three group minima. The other two primes in the two-prime groups are strictly larger than their group minima, and the third group minimum is at least  $d$ . Therefore the three primes other than  $g$  and  $d$  have product strictly larger than  $d^3$ . Hence

$$\rho^4 = \frac{d^4}{R} < \frac{d^4}{2d \cdot d^3} = \frac{1}{2}. \quad \square$$

**Theorem 9** (F24: type (2, 2, 1) upper bound). *For any type (2, 2, 1) triple,  $\rho^4 < 1/6$ , so  $\rho < 6^{-1/4} \approx 0.6389$ . The bound is strict for every finite triple. It would be approached by an infinite  $a = 6$  near-twin-prime family if such a family exists; that conditional sharpness*

is not claimed unconditionally. (Lean: `pasten.F29.221.rho4.lt.sixth`, which proves the general bound F29 covering all type (2, 2, 1) triples.)

*Proof.* Let  $g$  be the smallest group minimum and  $d$  the second-smallest; they lie in different groups. Let  $h > g$  be the other prime in  $g$ 's constituent, let  $q_2 > d$  be the other prime in  $d$ 's constituent, and let  $r \geq d$  be the single prime in the third group. Then

$$\rho^4 = \frac{d^4}{ghdq_2r} = \frac{d^3}{ghq_2r} < \frac{1}{6}$$

because  $g \geq 2$ ,  $h \geq 3$ ,  $q_2 \geq d + 1$ , and  $r \geq d$ , whence  $ghq_2r \geq 6d(d + 1) > 6d^2$ .  $\square$

**Theorem 10** (F23: types (0, 2, 3) and (0, 3, 2) upper bound). *For either type,  $\rho^4 < 1$ . A supremum equal to 1 would require an infinite near-equal prime family and is therefore conditional.*

*Proof.* Here  $a = 1$  and the two non-empty constituents each have at least two prime factors. Let  $d = nd$  be the larger of their two group minima, and let  $Y$  be the constituent whose minimum is  $d$ . Write  $Y = du$  with  $u > d$ ; since the primes are distinct,  $u \geq d + 2$ . The other constituent is either  $Y - 1$  or  $Y + 1$ , so it is at least  $Y - 1$ . Since  $R = \text{rad}(abc) = bc = Y(Y \pm 1)$ ,

$$R \geq d(d + 2)(d^2 + 2d - 1) > d^4.$$

Thus  $\rho^4 = d^4/R < 1$ .

In the special subfamily  $b = pq$  with  $3 \leq p < q$  and  $c = pq + 1$ , this becomes  $\rho^4 = p^3/(q(pq + 1))$ , the formula used in the finite computations.  $\square$

Type (2, 2, 1) has the strict upper bound  $6^{-1/4}$  (Theorem 9); conditional sharpness is approached by the  $a = 6$  family (see above). Types (0, 2, 3) and (0, 3, 2) have  $a = 1$  (empty  $P_a$ ), giving only two non-empty prime groups; their F10  $nd = \max(\min P_b, \min P_c)$  satisfies  $\rho < 1$  always (Theorem 10), with conditional sharpness at 1. The unbounded types (0, 1, 4) and (0, 4, 1) have  $\rho^4 \approx q^2 \rightarrow \infty$  analytically.

## F10: optimal crossing formula (general $\omega$ )

The key structural insight from **t21** completes the classification at all  $\omega$ :

**Theorem 11** (F10). *For any cross-group pair  $X \neq Y$  with  $X, Y \in \{P_a, P_b, P_c\}$ , the two-entry  $\varphi$ -vector with  $\varphi_p = 1$  ( $p \in X$ ) and  $\varphi_q = \pm 1$  ( $q \in Y$ , sign chosen to satisfy  $\sum \text{sign}_q \varphi_q = 0$ ) is always non-degenerate ( $S_b \neq S_a$ ). Its  $\psi$ -norm is  $\max(p, q)$ .*

*Consequently, the minimum non-degenerate norm satisfies*

$$\|\psi_{nd}\|_\infty = \min_{X \neq Y} \max(\min X, \min Y) = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\},$$

where  $\min \emptyset = \infty$ .

This formula corrects **t20**, which computed only the canonical  $P_a \times P_b$  crossing. The corrected classification of (3, 1, 1) is UNBOUNDED (not all-degenerate): non-degenerate vectors exist, but the minimum  $nd$  norm is  $\min(b, c)$ , which grows.

## E10/E11/E<sub>n</sub>: explicit minimum vectors and universal lower bound

Theorem 11 asserts an equality; we now give the explicit achieving vector and prove the matching lower bound, completing the characterisation.

**Theorem 12** (E10). *For type (1, 1, 1) with  $a = p \leq b = q \leq c = r$  all prime,  $p + q = r$ : the vector  $\psi = (p, -q, 0)$  lies in  $F(a, b)$ , satisfies  $W(\psi) = -2pq \neq 0$  (non-degenerate), and  $\|\psi\|_\infty = q = \text{nd}$ . (Lean: `pasten_E10_vec_in_lattice`, `nondeg`, `norm`.)*

**Theorem 13** (E11). *For type (1, 1, 2) with  $a = p$ ,  $b = q$ ,  $c = r_1 r_2$  ( $p, q, r_1, r_2$  distinct primes,  $p + q = r_1 r_2$ ,  $r_1 < r_2$ ): the vector  $\psi = (-p, 0, -r_1, 0)$  (i.e.,  $\varphi = (-1, 0, -1, 0)$  in the F8  $\varphi$ -coordinates) lies in  $F(a, b)$ , has  $W(\psi) = pq \neq 0$ , and  $\|\psi\|_\infty = \max(p, r_1) = \text{nd}$ . Verified numerically against all 310 triples with  $p \leq 19$ ,  $c \leq 500$  (`t46c`); Lean: `pasten_E11_vec_in_lattice`, `nondeg`, `norm`.*

**Theorem 14** (E<sub>n</sub>: universal lower bound). *For any squarefree coprime triple  $(a, b, c)$  with  $a + b = c$  and  $\omega(abc) = \omega$ , every non-degenerate  $\psi \in F(a, b)$  satisfies*

$$\|\psi\|_\infty \geq \text{nd} = \text{second smallest of } \{\min P_a, \min P_b, \min P_c\}.$$

*Proof.* By the Universal Divisibility Lemma (Theorem 6),  $\psi_\ell = \ell \varphi_\ell$  with  $\varphi_\ell \in \mathbb{Z}$ . Suppose for contradiction that  $\|\psi\|_\infty < \text{nd}$ . Then  $\ell |\varphi_\ell| \leq \|\psi\|_\infty < \text{nd}$ , so  $\varphi_\ell = 0$  whenever  $\ell \geq \text{nd}$ .

Write  $\text{nd} = \text{second smallest of } \{m_a, m_b, m_c\}$  with  $m_X = \min P_X$  (where  $\min \emptyset = \infty$ ). WLOG  $m_a = \min(m_a, m_b, m_c)$ ; then  $\text{nd} = \min(m_b, m_c)$ . Every  $\ell \in P_b$  satisfies  $\ell \geq m_b \geq \text{nd}$ , so  $\varphi_\ell = 0$ ; likewise every  $\ell \in P_c$  satisfies  $\ell \geq m_c \geq \text{nd}$ , so  $\varphi_\ell = 0$ . Hence  $S_b := \sum_{\ell \in P_b} \varphi_\ell = 0$  and  $S_c := \sum_{\ell \in P_c} \varphi_\ell = 0$ . The lattice constraint (F8  $\varphi$ -form) gives  $S_a + S_b = S_c$ , so  $S_a = 0$ . Non-degeneracy requires  $S_b \neq S_a$ , i.e.  $0 \neq 0$ —contradiction.  $\square$

If  $m_b$  is the global minimum, then  $\text{nd} = \min(m_a, m_c)$ ; every prime in  $P_a \cup P_c$  is at least  $\text{nd}$ , so  $\varphi$  vanishes on  $P_a \cup P_c$ . The constraint gives  $S_b = 0$ , contradicting  $S_b \neq S_a = 0$ . The case  $m_c$  global minimum is identical with  $S_c$  in place of  $S_b$ .  $\square$

Theorems 11 and 14 together give the complete characterisation:  $\min_{\psi \text{ nd}} \|\psi\|_\infty = \text{nd}$ , achieved and tight. Key-lemma Lean formalizations: `pasten_E10_lb_key` (Theorem 12) and `pasten_E11_lb_key` (Theorem 13). Numerical confirmation across all types:  $\omega = 4$  (`t47`),  $\omega = 5$  (`t48`),  $\omega = 6$  (`t49`)—all verified in  $\leq 1$  s.

## OB-13B: non-squarefree upper bound

The squarefree upper bound extends to non-squarefree triples with a  $v_{\max}$  factor, where  $v_{\max} = \max_{p \in P} v_p(abc)$ . For the finer non-squarefree formulas, and for the remainder of this non-squarefree discussion until the next squarefree classification theorem, we use the *valuation-coordinate lattice*

$$F_{\text{val}}(a, b) = \{\psi \in F(a, b) : p \mid \psi_p \text{ for every } p \in P\}.$$

Writing  $\psi_p = p \varphi_p$ , its constraint is

$$\sum_{p|a} v_p(a) \varphi_p + \sum_{p|b} v_p(b) \varphi_p - \sum_{p|c} v_p(c) \varphi_p = 0, \quad (1)$$

and its Wronskian condition is  $\sum_{p|b} \varphi_p \neq \sum_{p|a} \varphi_p$ . The inherited norm is  $\|\varphi\|_\infty = \max_{p \in P} p|\varphi_p|$ . Let  $\text{nd}_{\text{val}}$  denote the minimum non-degenerate norm on this sublattice; the symbol  $\text{nd}$  in these non-squarefree type formulas means  $\text{nd}_{\text{val}}$ . This convention is essential: for  $(1, 8, 9)$  the original lattice contains the non-degenerate vector  $\psi = (1, 2)$  of norm 2, whereas the valuation-coordinate minimum is 9. Thus exact valuation-lattice formulas do not automatically determine the original Pasten minimum.

**Theorem 15** (OB-13B: non-squarefree  $E_n$  upper bound). *For every coprime triple  $a+b=c$  (not necessarily squarefree) with  $c \leq 10^{12}$ , write  $\pi_1 < \pi_2 < \dots < \pi_{\omega^*}$  for the primes dividing  $R$ .*

$$\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}.$$

*For  $c > 10^{12}$  the bound also holds in all cases except possibly  $a = 1$ ,  $b = M^v$  with  $M \in \{6, 10\}$ ,  $v$  even,  $v \not\equiv 6 \pmod{12}$ , or  $M \geq 14$  and  $\pi_1, \pi_2 \in P_b$ ; for those sub-cases the proof is conditional on non-existence of  $R < c$  triples (see proof below).*

*Proof.* For each cross-group prime pair  $(p, q)$  with  $p \in P_g$ ,  $q \in P_{g'}$  ( $g \neq g'$ ) and  $g_0 = \gcd(v_p(g\text{-factor}), v_q(g'\text{-factor}))$ , define the vector  $\varphi^{(p,q)}$  by  $\varphi_p = v_q/g_0$ ,  $\varphi_q = -v_p/g_0$ , and  $\varphi_r = 0$  for all  $r \notin \{p, q\}$ . A direct computation shows  $\varphi^{(p,q)} \in F_{\text{val}}(a, b)$  (equation (1) is satisfied). The corresponding original-lattice vector  $\psi^{(p,q)}$  has  $\psi_p = p\varphi_p$  and  $\psi_q = q\varphi_q$ , and its valuation-coordinate Wronskian is  $\pm v_q/g_0 \neq 0$ , so it is non-degenerate. Its norm is  $\|\varphi^{(p,q)}\| = \max(p \cdot v_q/g_0, q \cdot v_p/g_0) \leq v_{\max} \cdot \max(p, q)$ .

Taking the minimum over all cross-group pairs and noting that when the optimal pair uses primes from groups  $G_1, G_2$ , we have  $\max(\min G_1, \min G_2) \leq \text{med}(m_a, m_b, m_c)$ :

$$\text{nd}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c).$$

It remains to show  $\pi_* := \text{med}(m_a, m_b, m_c) \leq R^{1/(\omega^*-1)}$ , i.e.  $\pi_*^{\omega^*-1} \leq R$ . We split into two cases.

**Case 1.** The two globally smallest primes  $\pi_1 < \pi_2$  in  $P$  lie in *different* groups (or  $P_a \neq \emptyset$  and a within-group vector on  $P_a$  or  $P_b$  uses primes with unequal valuations, giving a vector of norm  $\leq v_{\max} \cdot \pi_2$ ). Then  $\pi_* \leq \pi_2$ . The  $\omega^* - 1$  primes of  $R$  other than  $\pi_1$  all satisfy  $q_i \geq \pi_2$ , so their product  $\geq \pi_2^{\omega^*-1}$ , giving  $R \geq \pi_1 \cdot \pi_2^{\omega^*-1} \geq \pi_2^{\omega^*}$ , i.e.  $\pi_2 \leq R^{1/(\omega^*-1)}$ . ✓

**Case 2.**  $P_a = \emptyset$ ,  $P_b$  has equal valuations (every  $p \in P_b$  satisfies  $v_p(b) = v$ ), and  $\pi_1, \pi_2 \in P_b$ . Then within-group on  $P_b$  is degenerate ( $W = 0$ ), and the optimal  $P_b$ - $P_c$  cross-group pair gives  $\pi_* = \min(P_c) = \pi_3$  (the third-smallest prime globally). We need  $\pi_3^{\omega^*-1} \leq R$ . Write  $R/\pi_3 = \pi_1 \cdot \pi_2 \cdot \prod_{j \geq 4} \pi_j \geq 6 \cdot (\pi_3 + 2)^{\omega^*-3}$  (using  $\pi_1 = 2$ ,  $\pi_2 = 3$ , all  $\pi_j \geq \pi_3 + 2$  for  $j \geq 4$ ; there are  $\omega^* - 3$  such primes). For  $\pi_3 \leq 7$  and  $\omega^* \geq 4$ : we need  $\pi_3^{\omega^*-2} \leq 6 \cdot (\pi_3 + 2)^{\omega^*-3}$ . For  $\pi_3 = 7$ ,  $\omega^* = 4$ :  $7^2 = 49 \leq 6 \cdot 9 = 54$ . ✓ The ratio  $7^{\omega^*-2}/(6 \cdot 9^{\omega^*-3}) = (49/54) \cdot (7/9)^{\omega^*-4} \leq 49/54 < 1$  for all  $\omega^* \geq 4$ . ✓ (For  $\pi_3 = 5$ :  $5^2 = 25 \leq 6 \cdot 7 = 42$ , ratio  $(25/42) \cdot (5/7)^{\omega^*-4} < 1$ . ✓) For  $\pi_3 \geq 11$ : we analyse the two families of arising triples.

*Case 2a:*  $M = 10$ ,  $v$  odd ( $P_b = \{2, 5\}$ ,  $k = 2$ ). Since  $10 \equiv -1 \pmod{11}$ , we have  $11 \mid 1 + 10^v$  for all odd  $v$ , so  $\pi_3 = 11$ . Then  $R/\pi_3 = 10 \cdot \prod_{j \geq 4} \pi_j \geq 10 \cdot 13^{\omega^*-3}$ . For  $\omega^* = 4$ :  $10 \cdot 13 = 130 \geq 121 = 11^2$ . ✓ The ratio  $10 \cdot 13^{\omega^*-3}/11^{\omega^*-2} = (130/121) \cdot (13/11)^{\omega^*-4} > 1$  for all  $\omega^* \geq 4$ . ✓

*Case 2b:*  $M = 6$ ,  $v \equiv 6 \pmod{12}$  ( $P_b = \{2, 3\}$ ,  $k = 2$ ). Since  $5 \nmid c$  (as  $6 \equiv 1 \pmod{5}$ ),  $7 \nmid c$  ( $v$  even,  $6 \equiv -1 \pmod{7}$ ), and  $11 \nmid c$  ( $\text{ord}(6 \pmod{11}) = 10$ ,  $6^v \equiv -1 \pmod{11}$ ) requires

$v \equiv 5 \pmod{10}$ , which is odd), we have  $\pi_3 \geq 13$ . For  $\pi_3 = 13$  (requiring  $v \equiv 6 \pmod{12}$ ): the cyclotomic factor  $\Phi_{12}(6) = 6^4 - 6^2 + 1 = 1261 = 13 \times 97$  shows  $\text{ord}(6 \pmod{13}) = \text{ord}(6 \pmod{97}) = 12$  and  $6^6 \equiv -1$  modulo both 13 and 97. Moreover  $\text{ord}(6 \pmod{37}) = 4$  with  $6^2 \equiv -1 \pmod{37}$ , and  $v \equiv 6 \pmod{12}$  implies  $v \equiv 2 \pmod{4}$ , so  $37 \mid c$  too. Hence  $P_c \supseteq \{13, 37, 97\}$ ,  $|P_c| \geq 3$ ,  $\omega^* \geq 5$ ,  $R \geq 6 \cdot 13 \cdot 37 \cdot 97 = 279\,942$ . For  $\omega^* = 5$ :  $\pi_3^{\omega^*-1} = 13^4 = 28\,561 \leq 279\,942 = R$ . ✓ For  $\omega^* > 5$ : each extra  $P_c$  prime satisfies  $q \geq \pi_3 + 2 = 15$ , so  $R/\pi_3^{\omega^*-2}$  grows by a factor  $q/13 \geq 15/13 > 1$  per additional prime. ✓

For all other even- $v$  sub-cases ( $M = 6$ ,  $v$  even,  $v \not\equiv 6 \pmod{12}$ ;  $M = 10$ ,  $v$  even;  $M \geq 14$ ): exhaustive search (t96) over  $c \leq 10^{12}$  finds no  $R < c$  triple with  $\pi_1, \pi_2 \in P_b$  and  $|P_c| \geq 2$ . (A complete unconditional proof of non-existence in these sub-cases — showing that  $1 + M^v$  is squarefree enough to force  $R \geq c$  — requires a Wieferich-type prime-power analysis and is an open auxiliary obligation. The theorem therefore holds unconditionally for  $c \leq 10^{12}$  in these sub-cases and is conditional on non-existence of such triples for larger  $c$ .)

Verification summary: the complete list of  $R < c$  edge-case triples with  $\pi_1, \pi_2 \in P_b$  and  $c \leq 10^{12}$  is  $(1, 6^7, 279937)$  ( $\pi_3 = 7$ , proved above) and  $(1, 10^{11}, 10^{11} + 1)$  ( $\pi_3 = 11$ , Case 2a); both satisfy  $\pi_3^{\omega^*-1} \leq R$  with margins  $\times 100$  and  $\times 565\,000$  respectively. ✓  $\square$

*Remarks.* (i) For squarefree triples  $v_{\max} = 1$  and this recovers the Minkowski bound. (ii) The two-prime vector with GCD reduction can be sharpened to a multi-prime “balanced load” construction: for triples  $(1, b, q^k)$  with  $b$  squarefree and  $|P_b| \geq 2$ , the optimal vector distributes the constraint  $k$  across a subset  $T \subseteq P_b$ , achieving norm  $\max(q, k / \sum_{p_i \in T} p_i^{-1})$  — strictly below the pairwise bound. The exact formula  $\text{nd}_{\text{val}}(1, b) = \max(q, \min_T \lceil k/H(T) \rceil^*)$  with  $H(T) = \sum_{p_i \in T} 1/p_i$  is verified for all tested cases; no simple closed form exists for arbitrary valuation profiles, but  $\text{nd}_{\text{val}}(a, b)$  is computable exactly via LLL lattice reduction (Theorem 17 below, OB-13C). (iii) Numerical verification (t59, t65, t96): OB-13B holds for all 24 high-quality triples with quality  $> 1.15$  and  $c \leq 5000$ , with  $\text{nd}/(v_{\max} R^{1/(\omega^*-1)}) \leq 0.50$  in all cases. The tightest example is  $(1, 8, 9)$ :  $\text{nd} = 9$ ,  $v_{\max} = 3$ ,  $R = 6$ , ratio = 0.50. Case 2 edge-case examples:  $(1, 6^7, 279937)$  with  $P_b = \{2, 3\}$ ,  $P_c = \{7, 29, 197\}$ ,  $\pi_3 = 7$ ,  $R = 239946$ , norm =  $49 \leq 154.9 = v_{\max} R^{1/4}$  (ratio 0.32); and  $(1, 10^{11}, 10^{11} + 1)$  with  $P_b = \{2, 5\}$ ,  $P_c = \{11, 23, 4093, 8779\}$ ,  $\pi_3 = 11$ ,  $R = 90909090910$ , norm =  $121 \leq 1710.5$  (ratio 0.07). (iv) *Within-group construction.* When a group  $P_g$  contains two primes  $p, q$  with *distinct*  $p$ -adic valuations  $v_p \neq v_q$  of the  $g$ -factor, a within-group vector  $\varphi_p = v_q/g_0$ ,  $\varphi_q = -v_p/g_0$  (all other coordinates zero,  $g_0 = \gcd(v_p, v_q)$ ) satisfies the constraint (1) (both sides of 1 vanish for this vector) and has  $W(\varphi) = \pm(v_q - v_p)/g_0 \neq 0$ . This construction can give  $\text{nd}_{\text{val}}(a, b) < \text{med}(m_a, m_b, m_c)$ , i.e. below the squarefree  $E_n$  formula. Example:  $(a, b, c) = (72, 11, 83)$  with  $a = 2^3 \cdot 3^2$ ,  $P_a = \{2, 3\}$ ,  $P_b = \{11\}$ ,  $P_c = \{83\}$ . The within-group vector  $\varphi_2 = 2$ ,  $\varphi_3 = -3$  satisfies  $3 \cdot 2 + 2 \cdot (-3) = 0$  and  $W = 1 \neq 0$ , giving  $\text{nd}_{\text{val}}(72, 11) \leq 9 < 11 = \text{med}(m_a, m_b, m_c)$ . Thus the squarefree  $E_n$  lower bound  $\text{nd} \geq \text{med}$  does *not* extend to non-squarefree triples when groups have  $\geq 2$  primes with distinct valuations.

**Corollary 16** (OB-13B sharp for the valuation lattice). *Under the same hypotheses,*

$$\text{nd}_{\text{val}}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c),$$

where  $\text{med}$  denotes the second smallest of  $\{m_a, m_b, m_c\}$  (with  $m_g = \infty$  for empty groups).

*This bound is tight: equality holds at  $(a, b, c) = (1, 8, 9)$ , where  $v_{\max} = 3$ ,  $\text{med} = 3$ , and  $\text{nd}_{\text{val}} = 9$ .*

*Proof.* The proof of Theorem 15 constructs  $\text{nd}_{\text{val}}(a, b) \leq v_{\max} \cdot \text{med}(m_a, m_b, m_c)$  as an intermediate step (the Minkowski step that follows gives only a weaker bound). For  $(1, 8, 9)$ :  $P_a = \emptyset$ ,  $P_b = \{2\}$ ,  $P_c = \{3\}$ ;  $v_{\max} = 3$  (from  $v_2(8) = 3$ ),  $\text{med}(\infty, 2, 3) = 3$ . The optimal vector  $\varphi_2 = 2$ ,  $\varphi_3 = 3$  satisfies  $3 \cdot 2 = 2 \cdot 3$  (constraint) with  $W = 2 \neq 0$  and  $\text{norm} \max(2 \cdot 2, 3 \cdot 3) = 9 = 3 \cdot 3$ .  $\square$

### OB-13C: algorithmic exact computation of $\text{nd}_{\text{val}}(a, b)$

While no simple closed-form formula generalises the squarefree  $E_n$  expression  $\text{nd}_{\text{sq}} = \text{med}(m_a, m_b, m_c)$  to all non-squarefree triples, the exact value of  $\text{nd}_{\text{val}}(a, b)$  is computable by a finite deterministic lattice-reduction and enumeration procedure.

**Theorem 17** (OB-13C: valuation-lattice SVP algorithm). *Let  $a + b = c$  be a coprime triple with  $\omega^* = |P|$  distinct prime factors. Assume  $\omega^* \geq 2$ ; the unique  $\omega^* = 1$  triple  $(1, 1, 2)$  has rank 0 and no nonzero lattice vector. Write the constraint (1) as  $\sum_{i=1}^{\omega^*} \alpha_i \varphi_i = 0$  where  $\alpha_i = v_{p_i}(a)$  for  $p_i \in P_a$ ,  $\alpha_i = v_{p_i}(b)$  for  $p_i \in P_b$ , and  $\alpha_i = -v_{p_i}(c)$  for  $p_i \in P_c$ . The solution lattice  $L = \{\varphi \in \mathbb{Z}^{\omega^*} : \sum \alpha_i \varphi_i = 0\}$  has rank  $\omega^* - 1$ .*

1. *A  $\mathbb{Z}$ -basis for  $L$  is computed in  $O(\omega^{*2})$  operations by repeated application of the extended Euclidean algorithm to the coefficient vector  $(\alpha_i)$ .*
2. *Applying the LLL algorithm to this basis with the weighted inner product*

$$\langle u, v \rangle = \sum_i p_i^2 u_i v_i$$

*yields a reduced basis  $\{b_1, \dots, b_{\omega^*-1}\}$  with Gram-Schmidt vectors  $b_1^*, \dots, b_{\omega^*-1}^*$ .*

3. *Set  $U := v_{\max} \cdot \text{med}(m_a, m_b, m_c)$ , where  $m_g = \min P_g$  and empty group minima are interpreted as  $\infty$ . This is the unconditional two-prime witness bound constructed in the proof of Theorem 15. A standard depth-first enumeration over coordinates  $c_{\omega^*-1}, \dots, c_1$  with half-width*

$$R_j := \left\lceil \frac{U}{\|b_j^*\|_W} \right\rceil$$

*at level  $j$  (center updated from outer coordinates at each level; see proof) recovers the shortest non-degenerate valuation-coordinate vector, hence  $\text{nd}_{\text{val}}(a, b)$  exactly. The values  $R_j$  are computable after Step 2. A uniform bound is  $R_j \leq 2^{(\omega^*-2)/2} U / p_1$  where  $p_1 = \min P$ .*

*Proof.* The algorithm is implemented in `t64_omega4_lattice_svp.py` and checked against brute-force enumeration:  $\omega^* = 3$ : 13 test cases, all exact (`t63`);  $\omega^* = 4$ : 13 test cases, all exact (`t64`);  $\omega^* = 5$ : 6 test cases, all exact (`t64`).

*Step 1:* standard Smith normal form. The unimodular column transformation reducing  $[\alpha_1, \dots, \alpha_{\omega^*}]$  to  $[d, 0, \dots, 0]$  ( $d = \gcd(\alpha_i)$ ) supplies a  $\mathbb{Z}$ -basis for  $L$  as its last  $\omega^* - 1$  columns;

step 2 is standard LLL theory. The non-degeneracy constraint ( $W \neq 0$ ) eliminates the degenerate sublattice  $D \subset L$ .

*Step 3:* let  $\varphi^* = \sum_j c_j b_j$  be the nd-minimiser. The two-prime valuation-coordinate witness gives  $\|\varphi^*\|_W \leq \text{nd}_{\text{val}}(a, b) \leq U$ . Expanding in the Gram–Schmidt basis:  $\varphi^* = \sum_j \hat{c}_j b_j^*$  where  $\hat{c}_j = c_j + \sum_{i>j} c_i \mu_{i,j}$  (the GS coefficient). Cauchy–Schwarz in the weighted inner product gives  $|\hat{c}_j| \leq \|\varphi^*\|_W / \|b_j^*\|_W \leq U / \|b_j^*\|_W$ . Hence  $c_j = \hat{c}_j - T_j$  where  $T_j := \sum_{i>j} c_i \mu_{i,j}$  is determined by the outer coordinates  $c_{j+1}, \dots, c_{\omega^*-1}$ , and  $|c_j - (-T_j)| \leq R_j$ . The enumeration proceeds outermost-first ( $j = \omega^*-1$  down to 1); at level  $j$ , with all  $c_i$  ( $i > j$ ) fixed, the center  $-T_j$  is computed and  $c_j$  is searched in  $[-T_j - R_j, -T_j + R_j]$ . For  $j = \omega^*-1$  (outermost),  $T_{\omega^*-1} = 0$ , so the range is  $[-R_{\omega^*-1}, R_{\omega^*-1}]$ . Since  $\|b_j^*\|_W^2 \geq (1/2)^{j-1} \|b_1\|_W^2 \geq (1/2)^{j-1} p_1^2$  (Lovász condition with  $\delta = 3/4$ ; every non-zero  $L$ -vector has  $\|\cdot\|_W \geq p_1$ ), the uniform bound  $R_j \leq 2^{(j-1)/2} U / p_1 \leq 2^{(\omega^*-2)/2} U / p_1$  follows.  $\square$

### Universal lower bound and the $\omega^* = 2$ exact formula

The cross-group upper bound (Theorem 15) is complemented by a universal lower bound that holds for all coprime triples, squarefree or not.

**Lemma 18** (Single-prime constraint). *Let  $p \in P$  be any prime in  $\text{rad}(abc)$ . If  $\varphi \in F_{\text{val}}(a, b)$  satisfies  $\varphi_q = 0$  for all  $q \in P \setminus \{p\}$ , then  $\varphi_p = 0$ .*

*Proof.* Each prime lies in exactly one of  $P_a, P_b, P_c$ . With  $\varphi_q = 0$  for all  $q \neq p$ , constraint (1) reduces to  $v_p(\cdot) \varphi_p = 0$ , where  $v_p(\cdot) \geq 1$  since  $p \in P$ . Hence  $\varphi_p = 0$ .  $\square$

**Theorem 19** (Universal lower bound for  $\text{nd}_{\text{val}}$ ). *For every coprime triple  $a + b = c$ , let  $p_1 < p_2$  be the two smallest primes in  $P = \text{rad}(abc)$ . Then every non-degenerate vector in  $F_{\text{val}}(a, b)$  has norm at least*

$$p_2.$$

*Proof.* Let  $\varphi \in F_{\text{val}}(a, b)$  with  $W(\varphi) \neq 0$ . Suppose  $\|\varphi\| < p_2$ . For any  $p \geq p_2$  in  $P$ ,  $p |\varphi_p| \leq \|\varphi\| < p_2 \leq p$  forces  $|\varphi_p| < 1$ , so  $\varphi_p = 0$ . The only prime in  $P$  with  $p < p_2$  is  $p_1$ , so  $\varphi_q = 0$  for all  $q \neq p_1$ . By Lemma 18,  $\varphi_{p_1} = 0$  as well, giving  $\varphi = 0$ , which contradicts  $W(\varphi) \neq 0$ .  $\square$

**Remark 5.** *The bound is tight: for  $(a, b, c) = (1, 2, 3)$  with  $P = \{2, 3\}$ , the vector  $\varphi_2 = \varphi_3 = 1$  satisfies the constraint ( $1 \cdot 1 = 1 \cdot 1$ ) and has  $W = \varphi_2 = 1 \neq 0$ , giving  $\text{nd}_{\text{val}}(1, 2) = 3 = p_2$ .*

*For squarefree triples the  $E_n$  theorem gives  $\text{nd}_{\text{val}}(a, b) = \text{med}(m_a, m_b, m_c) \geq p_2$  (each group minimum is a prime in  $P$ ), so Theorem 19 is subsumed by  $E_n$  in that case. For non-squarefree triples it is the tightest universal lower bound: the within-group construction of Remark (iv) shows that  $\text{nd}_{\text{val}}(a, b)$  can fall strictly below  $\text{med}(m_a, m_b, m_c)$  (counterexample (72, 11, 83):  $\text{nd}_{\text{val}} = 9 > p_2 = 3$ ), so Theorem 19 is the right replacement for  $E_n$ 's lower bound.*

*Verified on 10 named non-squarefree examples including the within-group counterexample (72, 11, 83) (t67); the analytical proof above covers all triples.*

For triples with  $\omega^* = 2$  the lower bound is sharp and the exact value is determined by a one-dimensional lattice.



**Theorem 20** (Exact  $\text{nd}_{\text{val}}$  for  $\omega^* = 2$ ). *Let  $a + b = c$  be a coprime triple with  $\omega^* = 2$ , so  $P = \{p, q\}$  with  $p < q$ . Let  $v$  be the  $p$ -adic valuation and  $w$  the  $q$ -adic valuation of their respective factors in  $abc$ , and set  $g = \gcd(v, w)$ . Then*

$$\text{nd}_{\text{val}}(a, b) = \max\left(p \cdot \frac{w}{g}, q \cdot \frac{v}{g}\right) \geq q = p_2.$$

*Equality  $\text{nd}_{\text{val}} = q$  holds if and only if  $v = w = 1$ .*

*Proof.* The constraint (1) is a single equation  $v \varphi_p = w \varphi_q$  (with appropriate sign depending on the group memberships of  $p$  and  $q$ ). The solution lattice  $L$  has rank 1; its primitive generator is  $(\varphi_p, \varphi_q) = (w/g, \pm v/g)$ . All solutions are  $k \cdot (w/g, \pm v/g)$  for  $k \in \mathbb{Z}$ . Since  $W(\varphi) \neq 0$  iff the  $P_b$ -component is non-zero, this requires  $k \neq 0$ . The norm at  $|k| = 1$  is  $\max(p \cdot w/g, q \cdot v/g)$ , which is the minimum since the norm scales as  $|k|$ . The inequality  $\geq q$  follows from  $q \cdot v/g \geq q$  (as  $v/g \geq 1$ ) and  $p \cdot w/g \geq p \geq 2$ . Equality  $\text{nd}_{\text{val}} = q$  requires  $p \cdot w/g \leq q$  and  $q \cdot v/g = q$ , i.e.  $v = g$ ; combined with  $w/g = 1$  from the bound  $p \cdot w/g \leq q$  this gives  $v = w = g = 1$ .  $\square$

**Corollary 21** (Explicit  $\omega^* = 2$  formula). *Under the hypotheses of Theorem 20, with  $v = v_{p_1}$  and  $w = v_{p_2}$ :*

$$\text{nd}_{\text{val}}(a, b) = \frac{\max(p_1 \cdot v_{p_2}, p_2 \cdot v_{p_1})}{\gcd(v_{p_1}, v_{p_2})}.$$

*In particular,  $p_2 \leq \text{nd}_{\text{val}}(a, b) \leq p_2 \cdot v_{\max}$  with  $v_{\max} = \max(v_{p_1}, v_{p_2})$ .*

*Proof.* Substituting  $p = p_1$ ,  $q = p_2$ ,  $v = v_{p_1}$ ,  $w = v_{p_2}$ ,  $g = \gcd(v_{p_1}, v_{p_2})$  in Theorem 20 gives  $\text{nd}_{\text{val}}(a, b) = \max(p_1 w, p_2 v)/g = \max(p_1 v_{p_2}, p_2 v_{p_1})/\gcd(v_{p_1}, v_{p_2})$  directly. The lower bound  $\text{nd}_{\text{val}} \geq p_2$  follows from  $p_2 v/g \geq p_2 \geq p_2$  (as  $v \geq g$ ). The upper bound  $\text{nd}_{\text{val}} \leq p_2 v_{\max}$  follows from  $\max(p_1 w, p_2 v)/g \leq \max(p_2 w, p_2 v)/g = p_2 \cdot \max(v, w)/g \leq p_2 v_{\max}$  (since  $p_1 \leq p_2$  and  $g \geq 1$ ).  $\square$

**Corollary 22** (Complete norm spectrum for  $\omega^* = 2$ ). *Under the hypotheses of Theorem 20, the set of achievable norms of non-degenerate vectors in  $F(a, b)$  is exactly*

$$\Sigma(a, b) = \{k \cdot \text{nd}_{\text{val}}(a, b) : k \in \mathbb{Z}_{>0}\}.$$

*In particular, the successive minima satisfy  $\lambda_k = k \cdot \text{nd}_{\text{val}}(a, b)$  for all  $k \geq 1$ .*

*Proof.* The proof of Theorem 20 shows that all non-degenerate lattice vectors are  $m \cdot (w/g, \pm v/g)$  for  $m \in \mathbb{Z} \setminus \{0\}$ , with norm  $|m| \cdot \max(pw/g, qv/g) = |m| \cdot \text{nd}_{\text{val}}(a, b)$ . Thus every positive multiple of  $\text{nd}_{\text{val}}$  is achieved (take  $m = k$ ), and no other values occur.  $\square$

**Theorem 23** (Exact  $\text{nd}_{\text{val}}$  for type  $(k, 1, 1)$  non-squarefree triples). *Let  $a + b = c$  be a coprime triple with  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r\}$  (one prime per group,  $p < q < r$ ) and  $v_p(a) = k \geq 1$ ,  $v_q(b) = v_r(c) = 1$ . Then*

$$\text{nd}_{\text{val}}(a, b) = \min(r, qk).$$

*Proof. Upper bound.* The vector  $\varphi_p = 0, \varphi_q = \varphi_r = 1$  satisfies the constraint ( $k \cdot 0 + 1 = 1$ ) and has  $W = 1 - 0 = 1 \neq 0$ , giving norm  $r$ . The vector  $\varphi_p = -1, \varphi_q = k, \varphi_r = 0$  satisfies the constraint ( $k(-1) + k = 0$ ), has  $W = k - (-1) = k + 1 \neq 0$ , and norm  $\max(p, qk) = qk$  (as  $q > p$  and  $k \geq 1$  give  $qk \geq q > p$ ). Hence  $\text{nd}_{\text{val}}(a, b) \leq \min(r, qk)$ .

*Lower bound.* Let  $\varphi = (\varphi_p, \varphi_q, \varphi_r)$  be a non-degenerate vector.

- If  $\varphi_r \neq 0$ : then  $r|\varphi_r| \geq r$ , so  $\|\varphi\| \geq r \geq \min(r, qk)$ .
- If  $\varphi_r = 0$ : the constraint reduces to  $k\varphi_p + \varphi_q = 0$ , i.e.  $\varphi_q = -k\varphi_p$ . Non-degeneracy requires  $\varphi_q \neq \varphi_p$ , i.e.  $(k+1)\varphi_p \neq 0$ , so  $\varphi_p \neq 0$ . Then  $\|\varphi\| = \max(p|\varphi_p|, qk|\varphi_p|) = qk|\varphi_p| \geq qk \geq \min(r, qk)$ .

In both cases  $\|\varphi\| \geq \min(r, qk)$ .  $\square$

**Remark 6.** For  $k = 1$  (squarefree),  $\min(r, q) = q$  (as  $q < r$ ), recovering the  $E_n$  value. For  $k \geq 2$ :  $\text{nd}_{\text{val}} = qk$  when  $qk < r$  (the within-group effective valuation dominates), and  $\text{nd}_{\text{val}} = r$  otherwise. The crossover  $qk = r$  gives a transition between two regimes.

*Examples.* (4, 3, 7):  $p = 2, q = 3, r = 7, k = 2$ .  $\min(7, 6) = 6 = qk < r$  (dominant: valuation regime). (4, 7, 11):  $p = 2, q = 7, r = 11, k = 2$ .  $\min(11, 14) = 11 = r < qk$  (dominant: pairwise regime).

This gives a tight two-sided bound:  $\min(r, qk) \geq q = p_2$  (confirming Theorem 19), since  $qk \geq q$  and  $r > q$ .

**Theorem 24** (F34: Second successive minimum for type  $(k, 1, 1)$ ). *Under the hypotheses of Theorem 23, the second successive minimum is:*

(i) Valuation regime ( $qk < r, \text{nd}_{\text{val}} = qk$ ):

$$\lambda_2 = \min(r, 2qk).$$

(ii) Pairwise regime ( $qk \geq r, \text{nd}_{\text{val}} = r$ ):

$$\lambda_2 = \min\{N \in \{2r, qk, q(k-1)\} : N > r\}.$$

*Proof. Achievability.* We exhibit a non-degenerate vector for each candidate:

- $2r$ :  $\varphi = (0, 2, 2)$  (i.e.  $\varphi_p = 0, \varphi_q = \varphi_r = 2$ ). Constraint:  $k \cdot 0 + 2 = 2\checkmark$ .  $W = 2 - 0 = 2 \neq 0$ . Norm =  $\max(0, 2q, 2r) = 2r$ .
- $qk$ :  $\varphi = (1, -k, 0)$ . Constraint:  $k - k = 0\checkmark$ .  $W = -k - 1 \neq 0$ . Norm =  $\max(p, qk, 0) = qk$ .
- $q(k-1)$  (when  $q(k-1) > r$ ):  $\varphi = (1, 1-k, 1)$ . Constraint:  $k + (1-k) = 1\checkmark$ .  $W = (1-k) - 1 = -k \neq 0$ . Norm =  $\max(p, q(k-1), r) = q(k-1)$  (since  $q(k-1) > r > p$ ).

*Gap (no norm between  $\text{nd}_{\text{val}}$  and  $\lambda_2$ ).* Let  $M = \lambda_2$  and suppose  $\varphi \neq 0$  is non-degenerate with  $\text{nd}_{\text{val}} < \|\varphi\| < M$ .

(i) Valuation regime ( $\text{nd}_{\text{val}} = qk, M = \min(r, 2qk)$ ):  $r|\varphi_r| \leq \|\varphi\| < M \leq r$  forces  $\varphi_r = 0$ . Then  $k\varphi_p + \varphi_q = 0$ , so  $\|\varphi\| = qk|\varphi_p| \geq qk = \text{nd}_{\text{val}}$ . Equality gives  $|\varphi_p| = 1$ ; the next value is  $2qk \geq M$ .

(ii) Pairwise regime ( $\text{nd}_{\text{val}} = r$ ): write  $\varphi_p = t, \varphi_q = j - kt$  for integer  $t$ , where  $\varphi_r = j$  (any integer).

- $|j| = 0$ :  $\|\varphi\| = qk|t| \geq qk \geq M$ .
- $|j| \geq 2$ :  $\|\varphi\| \geq 2r \geq M$ .
- $|j| = 1$  (say  $j = 1$ ):  $\|\varphi\| = \max(p|t|, q|1 - kt|, r)$ . For  $\|\varphi\| < M$ , we need  $q|1 - kt| < M$ . Since  $M \leq q(k - 1)$  (or  $M = \min(2r, qk)$  when  $q(k - 1) \leq r$ ), one checks that  $|1 - kt| \leq k - 2$  (or  $|1 - kt| \leq k - 1$  resp.) forces  $kt \in [3 - k, k - 1]$ , leaving only  $t = 0$  (giving  $\|\varphi\| = r = \text{nd}_{\text{val}}$ ) as an integer solution.

In every case  $\|\varphi\| \geq M$ , so no norm lies strictly in  $(\text{nd}_{\text{val}}, M)$ .  $\square$

**Remark 7.** *The formula specialises cleanly: for  $k = 1$  (type  $(1, 1, 1)$ , always valuation),  $\lambda_2 = \min(r, 2q) = r$  since  $r = q + 2 < 2q$  for twin primes  $q \geq 3$  (recovering Theorem 31). For  $k = 2$  pairwise ( $r \leq 2q$ ): since  $q(k - 1) = q \leq r$ , one has  $\lambda_2 = \min(2r, 2q)$ . T87 verifies Theorem 24 on 55 triples (four base primes  $p \in \{2, 3, 5, 7\}$ ,  $k \leq 9$ , adaptive bound), zero failures.*

**Theorem 25** (Exact  $\text{nd}_{\text{val}}$  for type  $(k, m, 1)$  non-squarefree triples). *Let  $a + b = c$  be a coprime triple with  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r\}$  and  $v_p(a) = k \geq 1$ ,  $v_q(b) = m \geq 1$ ,  $v_r(c) = 1$ ,  $p < q < r$ . Let  $g = \gcd(k, m)$ . Then*

$$\text{nd}_{\text{val}}(a, b) = \min\left(r, \max\left(\frac{pm}{g}, \frac{qk}{g}\right)\right).$$

*Proof. Upper bound.* The vector  $\varphi_p = -m/g$ ,  $\varphi_q = k/g$ ,  $\varphi_r = 0$  satisfies the constraint  $(k \cdot (-m/g) + m \cdot (k/g) = 0)$ , has  $W = k/g + m/g = (k + m)/g \neq 0$ , and norm  $\max(pm/g, qk/g)$ . Hence  $\text{nd}_{\text{val}}(a, b) \leq \max(pm/g, qk/g)$ . When  $\max(pm/g, qk/g) \leq r$  this gives the desired upper bound  $\text{nd}_{\text{val}}(a, b) \leq \min(r, \max(pm/g, qk/g))$ . When  $\max(pm/g, qk/g) > r$  (pairwise regime, which forces  $m = 1$  for actual triples  $a = p^k$ ,  $b = q$ ,  $c = p^k + q$  prime), the upper bound  $\text{nd}_{\text{val}}(a, b) \leq r$  follows from Theorem 23 (the vector  $\varphi_p = 0$ ,  $\varphi_q = 1$ ,  $\varphi_r = 1$  has norm  $r$  when  $m = 1$ ).

*Lower bound —  $\varphi_r = 0$  case.* The constraint reduces to  $k\varphi_p + m\varphi_q = 0$ . Since  $\gcd(m/g, k/g) = 1$ , the integer solutions are  $\varphi_p = -(m/g)t$ ,  $\varphi_q = (k/g)t$  for  $t \in \mathbb{Z}$ . Non-degeneracy  $W = (k/g + m/g)t \neq 0$  forces  $t \neq 0$ , giving norm  $\max(pm/g, qk/g) \cdot |t| \geq \max(pm/g, qk/g)$ .

*Lower bound —  $\varphi_r \neq 0$  case.* Then  $r|\varphi_r| \geq r$ , so  $\|\varphi\| \geq r$ .

Combining both cases,  $\text{nd}_{\text{val}}(a, b) \geq \min(r, \max(pm/g, qk/g))$ .  $\square$

**Remark 8.** *Theorem 23 is the special case  $m = 1$  (so  $g = 1$ ,  $\max(p, qk) = qk$  since  $qk > p$  for  $k \geq 1$ ,  $q > p$ ). When  $k = m$  (equal valuations),  $g = k$  and  $\max(p, q) = q$ , giving  $\text{nd}_{\text{val}} = q = p_2$ : equal valuations always realise the lower bound. When  $k = m = 1$  (squarefree),  $\text{nd}_{\text{val}} = q$ , recovering  $E_n$ .*

**Corollary 26** (Valuation regime is independent of  $n$ ). *Let  $a = p^k$ ,  $b = q^m$ ,  $c = r^n$  be coprime with  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r\}$  and  $p < q < r$ . Set  $g = \gcd(k, m)$  and  $N_0 = \max(pm/g, qk/g)$ . If  $N_0 \leq r$  (valuation regime), then*

$$\text{nd}_{\text{val}}(a, b) = N_0 = \max\left(\frac{pm}{g}, \frac{qk}{g}\right),$$

regardless of the exponent  $n = v_r(c)$ .

*Proof.* The witness  $\varphi_p = -m/g$ ,  $\varphi_q = k/g$ ,  $\varphi_r = 0$  satisfies  $k(-m/g) + m(k/g) = 0 = n \cdot 0$  for any  $n$ , has  $W = (k + m)/g \neq 0$ , and achieves norm  $N_0$ . The lower bound argument of Theorem 25 is verbatim: when  $\varphi_r = 0$  the constraint reduces to  $k\varphi_p + m\varphi_q = 0$ , giving norm  $\geq N_0$ ; when  $\varphi_r \neq 0$ , norm  $\geq r \geq N_0$ . Neither case involves  $n$ .  $\square$

**Theorem 27** (F35: Second successive minimum for type  $(k, m, 1)$  in the valuation regime). *Let  $a + b = c$  be a coprime triple with  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r\}$ ,  $v_p(a) = k \geq 2$ ,  $v_q(b) = m \geq 2$ ,  $v_r(c) = 1$ ,  $p < q < r$ . Set  $g = \gcd(k, m)$  and  $N_0 = \max(pm/g, qk/g)$ . In the valuation regime ( $N_0 < r$ ), the second successive minimum satisfies*

$$\lambda_2(a, b) = \min(r, 2N_0),$$

*with the convention that  $r$  is achievable only when  $g = 1$  (coprime exponents); when  $g \geq 2$ , the  $|\varphi_r| = 1$  branch has no integer solutions and  $\lambda_2 = 2N_0$  (which equals  $\min(r, 2N_0)$  in all practical cases since  $2N_0 \leq r$  holds whenever  $g \geq 2$  for prime triples with  $k, m \geq 2$ ).*

*Proof. Upper bound.* The vector  $\varphi_p = -2m/g$ ,  $\varphi_q = 2k/g$ ,  $\varphi_r = 0$  lies in the lattice ( $k(-2m/g) + m(2k/g) = 0$ ), has  $W = 2k/g + 2m/g = 2(k + m)/g \neq 0$ , and achieves norm  $2N_0 > N_0 = \text{nd}_{\text{val}}(a, b)$ . When  $g = 1$  and  $r \leq 2N_0$ : the minimal Bezout solution  $(k_0, m_0)$  of  $kk_0 + mm_0 = 1$  satisfies  $|k_0| \leq m/2$  and  $|m_0| \leq k/2$ , so  $p|k_0| \leq N_0/2 < r$  and  $q|m_0| \leq N_0/2 < r$ ; thus the vector  $(\varphi_p, \varphi_q, \varphi_r) = (k_0, m_0, 1)$  has norm  $\max(p|k_0|, q|m_0|, r) = r$ . Hence  $\lambda_2 \leq \min(r, 2N_0)$ .

*Lower bound.* Split every non-minimizing vector by its  $\varphi_r$  value:

- $\varphi_r = 0$ : the constraint reduces to  $k\varphi_p + m\varphi_q = 0$ , whose integer solutions are  $(\varphi_p, \varphi_q) = (-(m/g)t, (k/g)t)$  for  $t \in \mathbb{Z}$ . Since  $\gcd(m/g, k/g) = 1$ , the generator is primitive, and norm  $= |t|N_0 \geq 2N_0$  for  $|t| \geq 2$ .
- $|\varphi_r| \geq 1$ : norm  $\geq r \cdot |\varphi_r| \geq r \geq \min(r, 2N_0)$  (using  $N_0 < r$ ).

Every non-minimizing vector therefore has norm  $\geq \min(r, 2N_0)$ , giving  $\lambda_2 \geq \min(r, 2N_0)$ .  $\square$

**Remark 9.** *Theorem 27 is the direct generalization of the valuation-regime case of Theorem 24 (which has  $m = 1$ ,  $g = 1$ ,  $N_0 = qk$ ). The pairwise regime ( $N_0 \geq r$ ) is vacuous for  $k, m \geq 2$ : one would need  $\max(pm/g, qk/g) \geq p^k + q^m$ , which requires  $q \cdot \max(k, m)/g \geq q^m$ , i.e.  $\max(k, m)/g \geq q^{m-1}$ ; impossible for  $m \geq 2$  and  $q \geq 3$ . T88b verifies Theorem 27 on 40 triples ( $p \leq 7$ ,  $k, m \in \{2, \dots, 9\}$ ,  $a, b \leq 2000$ , zero failures).*

**Remark 10** (Mixed witnesses in the pairwise regime). *Theorems 25 and 26 require  $p_c = r$  to be the largest prime. When the group- $c$  prime is not the largest, neither the valuation-regime formula nor the three-pure formula  $\min(N_0, N_1, N_2)$  is guaranteed correct: a fully mixed (no-zero-coordinate) witness can achieve a smaller norm than any pure branch.*

*The canonical example is  $(5, 27, 32)$ , i.e.  $p_a = 5 > p_b = 3 > p_c = 2$ ,  $k = 1$ ,  $m = 3$ ,  $n = 5$ . The three pure branches give  $N_0 = N_1 = 15$ ,  $N_2 = 25$ , so  $\min(N_0, N_1, N_2) = 15$ . But the mixed vector  $(\varphi_{p_a}, \varphi_{p_b}, \varphi_{p_c}) = (-1, 2, 1)$  satisfies  $1 \cdot (-1) + 3 \cdot 2 = 5 \cdot 1$ , has  $W = 3 \neq 0$ , and achieves norm  $\max(5, 6, 2) = 6$ . Script T70 verifies  $\text{nd}_{\text{val}}(5, 27) = 6$  by exhaustive search. In such cases the exact  $\text{nd}$  is computed by the LLL algorithm of Theorem 17.*

**Theorem 28** (Exact nd in pairwise regime via Bezout). *Let  $a + b = c$  be coprime with  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r\}$  (one prime per group), exponents  $k, m, n$ , and let  $p_L = \max(p, q, r)$ . Write the lattice constraint as  $k_p\varphi_p + k_q\varphi_q + k_r\varphi_r = 0$  where  $k_p = k$ ,  $k_q = m$ ,  $k_r = -n$ . Let*

$N = \text{minimum norm of the } \varphi_{p_L}=0 \text{ sub-problem,}$

$$B = \min \left\{ \max_{p_i \neq p_L} p_i |\varphi_i| \mid k_{p_L} \cdot (\pm 1) + \sum_{p_i \neq p_L} k_{p_i} \varphi_i = 0, W \neq 0 \right\}$$

*be the minimum weighted norm over all  $\varphi_{p_L} = \pm 1$  solutions (set  $B = \infty$  if none exist, i.e.  $\gcd(k_{\text{other1}}, k_{\text{other2}})$  does not divide  $k_{p_L}$ ).*

*In the pairwise regime  $N > p_L$ :*

- (i)  $\text{nd}_{\text{val}}(a, b) \geq p_L$  (lower bound, since  $\varphi_{p_L} = 0$  gives norm  $\geq N > p_L$  and  $\varphi_{p_L} \neq 0$  gives norm  $\geq p_L$ );
- (ii)  $\text{nd}_{\text{val}}(a, b) = p_L$  if and only if  $B \leq p_L$ ;
- (iii)  $\text{nd}_{\text{val}}(a, b) = \min(N, B)$  when  $B > p_L$  (the exact value is either the pure-branch norm  $N$  or the Bezout-family minimum  $B$ , whichever is smaller).

*In particular, the unified exact formula for all one-prime-per-group triples (both regimes) is*

$$\text{nd}_{\text{val}}(a, b) = \min(N, \max(p_L, B)).$$

*Proof.* Lower bound (i): already noted above ( $\varphi_{p_L} = 0$  forces norm  $\geq N$ ;  $\varphi_{p_L} \neq 0$  forces  $p_L |\varphi_{p_L}| \geq p_L$ ).

For (ii): if  $B \leq p_L$ , the Bezout witness  $\varphi_{p_L} = \pm 1$  with the optimal partner achieves norm  $\max(p_L, B) = p_L$ . Conversely, if  $\text{nd}_{\text{val}} = p_L$ , then a minimising vector  $\varphi$  has norm  $p_L$ , forcing  $|\varphi_{p_L}| \leq 1$ ; the  $\varphi_{p_L} = 0$  case gives norm  $\geq N > p_L$ , so  $|\varphi_{p_L}| = 1$  and that vector is a Bezout witness with max other-prime norm  $\leq p_L$ , i.e.  $B \leq p_L$ .

For (iii):  $\text{nd}_{\text{val}} = \min(\text{norm from } \varphi_{p_L} = 0, \text{ norm from } |\varphi_{p_L}| = 1)$  since  $|\varphi_{p_L}| \geq 2$  gives norm  $\geq 2p_L > \min(N, B)$ . The first gives  $N$  and the second gives  $\max(p_L, B) = B$  (since  $B > p_L$ ), so  $\text{nd}_{\text{val}} = \min(N, B)$ . The unified formula  $\min(N, \max(p_L, B))$  covers both sub-cases and the valuation regime ( $N \leq p_L$ , where it reduces to  $N$ ).  $\square$

**Remark 11.** *Script T72 verifies Theorem 28 on all 229 one-prime-per-group pairwise-regime triples in  $[1, 600]$ : the Bezout condition  $B \leq p_L$  correctly predicts  $\text{nd} = p_L$  in all 223 such cases, and fails in the 6 exceptions (e.g.  $(5, 27)$ ,  $(7, 25)$ ,  $(169, 343)$ ) where  $B > p_L$  and  $\text{nd} = \min(N, B)$ .*

**Remark 12** (T73: high-quality triples). *Script T73 scans one-prime-per-group triples  $p_a^k + p_b^m \leq 5 \times 10^5$  for quality  $> 1$  and finds exactly 10, listed by decreasing quality:*

| <i>Triple</i> | <i>quality</i> | <i>nd</i> | <i>p<sub>L</sub></i> | <i>nd vs p<sub>L</sub></i> | <i>branch</i>                                 |
|---------------|----------------|-----------|----------------------|----------------------------|-----------------------------------------------|
| (3, 125)      | 1.427          | 9         | 5                    | >                          | pairwise, $B = N_0 = 9$                       |
| (7, 32761)    | 1.327          | 105       | 181                  | <                          | $\text{val}(\varphi_{p_L} = 0)$ , $N_2 = 105$ |
| (13, 243)     | 1.273          | 13        | 13                   | =                          | pairwise, $B = 9 \leq 13$                     |
| (169, 343)    | 1.199          | 21        | 13                   | >                          | pairwise, $B = \infty$ , $N = 21$             |
| (32, 49)      | 1.176          | 7         | 7                    | =                          | pairwise, $B = 6 \leq 7$                      |
| (751, 130321) | 1.149          | 323       | 751                  | <                          | $\text{val}(\varphi_{p_L} = 0)$ , $N_1 = 323$ |
| (139, 2048)   | 1.143          | 33        | 139                  | <                          | $\text{val}(\varphi_{p_L} = 0)$ , $N_1 = 33$  |
| (128, 4913)   | 1.095          | 71        | 71                   | =                          | pairwise, $B = 71 = p_L$                      |
| (4, 121)      | 1.027          | 10        | 11                   | <                          | $\text{val}(\varphi_{p_L} = 0)$ , $N_2 = 10$  |
| (5, 27)       | 1.019          | 6         | 5                    | >                          | pairwise, $B = 6 > p_L$                       |

All 10 are correctly classified by Theorem 28 and Corollary 26. Four triples satisfy  $nd < p_L$ : the winning branch is the  $\text{val}(\varphi_{p_L} = 0)$  case, which zeroes the largest prime-group coordinate (rather than the  $c$ -group as in the standard valuation-regime). Note that quality and  $nd$  are structurally independent:  $nd$  is determined by  $(p_a, p_b, p_c, k, m)$  while quality depends on  $n$ .

**Corollary 29** (Minimum depends only on the smallest  $c$ -prime). *Let  $a + b = c$  be coprime with  $P_a = \{p\}$ ,  $P_b = \{q\}$  (one prime each), and  $P_c$  any finite set of primes with  $r_1 = \min P_c > q > p$ . Set  $g = \gcd(k, m)$  where  $k = v_p(a)$ ,  $m = v_q(b)$ . Then*

$$\text{nd}_{\text{val}}(a, b) = \min\left(r_1, \max\left(\frac{pm}{g}, \frac{qk}{g}\right)\right).$$

In particular, the formula depends on  $c$  only through  $r_1 = \min P_c$ : the higher primes in  $P_c$  and their valuations are irrelevant.

*Proof.* Let  $N_0 = \max(pm/g, qk/g)$ . *Lower bound.* If all  $P_c$ -coordinates of  $\varphi$  are zero, the constraint reduces to the  $\omega^* = 2$  problem on  $\{p, q\}$ , giving norm  $\geq N_0$  (identical to the  $\varphi_r = 0$  case in Theorem 25). If at least one coordinate  $\varphi_r$  with  $r \in P_c$  is nonzero, then  $r|\varphi_r| \geq r \geq r_1$ . Hence  $\text{nd}_{\text{val}}(a, b) \geq \min(r_1, N_0)$ .

*Upper bound.* The  $P_c$ -zero witness (from Theorem 25) achieves norm  $N_0$ . The pairwise-regime bound  $\text{nd}_{\text{val}} \leq r_1$  follows from Theorem 25 applied to the sub-triple  $(a, b, r_1^{v_{r_1}(c)})$  (which has  $P_c = \{r_1\}$ , the same lower bound, and the same  $N_0$ ).  $\square$

The next group of higher-dimensional type formulas has a common proof boundary: the displayed witnesses and the cases with a zero coordinate are proved, but the all-nonzero case has only finite verification. They are therefore printed as candidate formulas rather than theorems.

**Candidate Formula 1** (Exact  $\text{nd}_{\text{val}}$  for  $\omega^* = 4$  with two  $c$ -group primes). *Let  $a + b = c$  be coprime with  $P_a = \{p\}$ ,  $P_b = \{q\}$ , and  $P_c = \{r, s\}$  (all four primes distinct), with  $k = v_p(a)$ ,  $m = v_q(b)$ ,  $j = v_r(c)$ ,  $n = v_s(c)$ . Define*

$$B_x := \min_{\substack{j\varphi_r + n\varphi_s = x \\ \varphi_r, \varphi_s \in \mathbb{Z}}} \max(r|\varphi_r|, s|\varphi_s|) \quad (= \infty \text{ if } \gcd(j, n) \nmid x).$$

Then

$$\text{nd}_{\text{val}}(a, b) = \min\left(\text{nd}_3(p^k, q^m, r^j), \quad \text{nd}_3(p^k, q^m, s^n), \quad \max(q, B_m), \quad \max(p, B_k)\right),$$

where  $\text{nd}_3(\cdot)$  denotes the one-prime-per-group formula from Theorem 28.

*Evidence. Upper bound.* Each branch gives an explicit non-degenerate witness:

- $\varphi_s = 0$  branch: the optimal vector for the sub-triple  $(p^k, q^m, r^j)$  with  $P_c = \{r\}$  has norm  $\text{nd}_3(p^k, q^m, r^j)$  and satisfies  $k\varphi_p + m\varphi_q = j\varphi_r$  with  $W = \varphi_q - \varphi_p \neq 0$ .
- $\varphi_r = 0$  branch: analogously for  $(p^k, q^m, s^n)$ .
- $\varphi_p = 0, \varphi_q = \pm 1$  branch: set  $j\varphi_r + n\varphi_s = \pm m$  with the pair  $(\varphi_r, \varphi_s)$  achieving  $B_m$ ; then  $\varphi_p = 0$  satisfies the constraint,  $W = \varphi_q \neq 0$ , and the norm is  $\max(q, B_m)$ .
- $\varphi_q = 0, \varphi_p = \pm 1$  branch: analogously, achieving  $\max(p, B_k)$ .

*Lower bound.* If  $\varphi_s = 0$ : norm  $\geq \text{nd}_3(p^k, q^m, r^j)$  by Theorem 28 applied to the three-prime sub-problem. If  $\varphi_r = 0$ : analogously, norm  $\geq \text{nd}_3(p^k, q^m, s^n)$  by Theorem 28. If  $\varphi_p = 0$  and  $\varphi_q \neq 0$ : norm  $\geq q|\varphi_q| \geq q$ , and the pair  $(\varphi_r, \varphi_s)$  satisfies  $j\varphi_r + n\varphi_s = m\varphi_q$ , giving norm  $\geq \max(q, B_{m|\varphi_q|}) \geq \max(q, B_m)$ . If  $\varphi_q = 0$  and  $\varphi_p \neq 0$ : analogously  $\geq \max(p, B_k)$ . The case where all four coordinates are nonzero is verified computationally (script T74, 431 cases in  $[1, 120)$ , 0 failures): no such witness achieves a smaller norm than the minimum of the four branches.  $\square$

**Remark 13.** Script T74 verifies Candidate Formula 1 on all 431 such triples in  $[1, 120)$  (initial run: 262 triples in  $[1, 80)$ ): 0 failures in both ranges. The  $\varphi_q = 0$  branch is the minimum in 25 cases (e.g.  $(37, 71)$ :  $p = 37, q = 71, c = 2^2 \cdot 3^3, \text{nd}_{\text{val}} = 37 = \max(37, B_1 = 3)$ ); the  $\varphi_p = 0$  branch wins in 5 cases. The two  $\varphi_{c\text{-prime}} = 0$  branches suffice in 223/262 cases.

**Candidate Formula 2** (Exact  $\text{nd}_{\text{val}}$  for type  $(2, 1, 1)$  triples). Let  $a + b = c$  be coprime with  $P_a = \{p, q\}$  ( $p < q$ , exponents  $k_1 = v_p(a), k_2 = v_q(a)$ ),  $P_b = \{r\}$  (exponent  $m$ ),  $P_c = \{s\}$  (exponent  $n$ ), all four primes distinct. Define the  $P_a$ - $P_b$  interaction norm

$$N_{s=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|) : k_1\varphi_p + k_2\varphi_q + m\varphi_r = 0, \varphi_r \neq \varphi_p + \varphi_q\}$$

(zeroing the  $P_c$ -coordinate; subsumes the within-group norm and the two  $P_a$ - $P_b$  pairwise norms), and the  $P_a$ - $P_c$  interaction norm

$$N_{r=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, s|\varphi_s|) : k_1\varphi_p + k_2\varphi_q = n\varphi_s, \varphi_s \neq 0, \varphi_p + \varphi_q \neq 0\}$$

(zeroing  $P_b$ ; subsumes the two  $P_a$ - $P_c$  pairwise norms). Let  $\text{nd}_3(p^{k_1}, r^m, s^n)$  and  $\text{nd}_3(q^{k_2}, r^m, s^n)$  denote the exact  $\text{nd}_{\text{val}}$  values for the  $\omega^* = 3$  sub-problems at primes  $(p, r, s)$  and  $(q, r, s)$  with exponent vectors  $(k_1, m, n)$  and  $(k_2, m, n)$  (computed via Theorem 28; each subsumes the corresponding pairwise norms and the  $P_b$ - $P_c$  pairwise norm).

Then

$$\text{nd}_{\text{val}}(a, b) = \min\left(N_{s=0}, N_{r=0}, \text{nd}_3(p^{k_1}, r^m, s^n), \text{nd}_3(q^{k_2}, r^m, s^n)\right).$$

*Evidence. Upper bound.* Each branch provides a constructive non-degenerate witness:  $N_{s=0}$  and  $N_{r=0}$  use a Bezout construction on the  $P_a$ -sub-lattice (analogous to Candidate Formula 1); the  $\omega^* = 3$  branches use Theorem 28 directly.

*Lower bound.* Any non-degenerate  $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s)$  with  $k_1\varphi_p + k_2\varphi_q + m\varphi_r = n\varphi_s$  and  $W = \varphi_r - (\varphi_p + \varphi_q) \neq 0$  satisfies at least one of:

- $\varphi_s = 0$ : norm  $\geq N_{s=0}$  by definition.
- $\varphi_r = 0, \varphi_s \neq 0$ : norm  $\geq N_{r=0}$  by definition.
- $\varphi_q = 0$ :  $\varphi$  lies in the  $\omega^* = 3$  sub-lattice for  $(p^{k_1}, r^m, s^n)$ , so norm  $\geq \text{nd}_3(p^{k_1}, r^m, s^n)$ .
- $\varphi_p = 0$ : analogously norm  $\geq \text{nd}_3(q^{k_2}, r^m, s^n)$ .
- All four coordinates non-zero: verified computationally (script T77, 672 triples in  $[4, 150) \times [1, 150)$ , 0 failures).

The first four cases cover all vectors with at least one zero coordinate; note that a vector may fall in more than one case, but the minimum suffices.  $\square$

**Remark 14.** *The four branches correspond to zeroing one of the four prime-group slots.  $N_{s=0}$  subsumes the within-group norm, all  $P_a$ - $P_b$  pairwise norms, and any vector with  $\varphi_r = \varphi_s = 0$ .  $N_{r=0}$  subsumes all  $P_a$ - $P_c$  pairwise norms. Each  $\omega^* = 3$  branch subsumes the  $P_b$ - $P_c$  pairwise norm and the remaining two pairwise norms for its  $P_a$ -prime. Script T77 (672 triples, 0 failures) confirms that no all-nonzero vector can improve on this minimum:  $N_{s=0}$  is the minimum in 582 cases (87%),  $\text{nd}_3(p^{k_1}, r^m, s^n)$  in 378,  $N_{r=0}$  in 70,  $\text{nd}_3(q^{k_2}, r^m, s^n)$  in 172.*

**Candidate Formula 3** (Exact  $\text{nd}_{\text{val}}$  for type  $(1, 2, 1)$  triples). *Let  $a + b = c$  be coprime with  $P_a = \{p\}$  (exponent  $k$ ),  $P_b = \{q, r\}$  ( $q < r$ , exponents  $j = v_q(b)$ ,  $m = v_r(b)$ ),  $P_c = \{s\}$  (exponent  $n$ ), all four primes distinct. Define the  $P_a$ - $P_b$  interaction norm*

$$N_{s=0} := \min\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|) : k\varphi_p + j\varphi_q + m\varphi_r = 0, (\varphi_q + \varphi_r) \neq \varphi_p\}$$

*(zeroing the  $P_c$ -coordinate; subsumes the within-group  $P_b$  norm and the two  $P_a$ - $P_b$  pairwise norms), and the  $P_b$ - $P_c$  interaction norm*

$$N_{p=0} := \min\{\max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : j\varphi_q + m\varphi_r = n\varphi_s, \varphi_q + \varphi_r \neq 0\}$$

*(zeroing  $P_a$ ; subsumes the two  $P_b$ - $P_c$  pairwise norms). Let  $\text{nd}_3(p^k, q^j, s^n)$  and  $\text{nd}_3(p^k, r^m, s^n)$  denote the  $\omega^* = 3$  sub-problems (Theorem 28; each subsumes the  $P_a$ - $P_c$  pairwise norm and the corresponding  $P_b$ - $P_c$  norm). Then*

$$\text{nd}_{\text{val}}(a, b) = \min\left(N_{s=0}, N_{p=0}, \text{nd}_3(p^k, q^j, s^n), \text{nd}_3(p^k, r^m, s^n)\right).$$

*Evidence.* The Wronskian is  $W = (\varphi_q + \varphi_r) - \varphi_p$  and the constraint is  $k\varphi_p + j\varphi_q + m\varphi_r = n\varphi_s$ . This has the same algebraic structure as Candidate Formula 2 with  $P_a$  and  $P_b$  exchanged: relabelling  $p \leftrightarrow r$ ,  $k_1 \leftrightarrow m$ ,  $k_2 \leftrightarrow j$ , and identifying the single-prime group as  $P_a = \{p\}$  maps the four-branch formula of Candidate Formula 2 verbatim to the formula above. The subsumption claims are verified by setting one or two coordinates to zero in the respective minimisation problems.  $\square$



**Remark 15.** Script T78\_formula verifies Candidate Formula 3 on all 1043 type (1, 2, 1) triples with  $a, b \leq 200$ : 0 failures. Branch win counts (ties included):  $N_{s=0}$  in 935 cases (90%),  $\text{nd}_3(p^k, q^j, s^n)$  in 567,  $N_{p=0}$  in 395,  $\text{nd}_3(p^k, r^m, s^n)$  in 265. One spot-check case ( $a = 2^7$ ,  $b = 3 \cdot 23$ ,  $c = 197$ ) requires  $|\varphi_3| = 7$  in the optimal witness via  $N_{s=0}$  ( $\text{nd}_{\text{val}} = 21$ ).

The same recursive strategy extends to  $\omega = 5$ .

**Candidate Formula 4** (Exact  $\text{nd}_{\text{val}}$  for type (2, 2, 1) triples). Let  $a + b = c$  be coprime with  $P_a = \{p, q\}$  (exponents  $k_1, k_2$ ),  $P_b = \{r, s\}$  (exponents  $j_1, j_2$ ),  $P_c = \{u\}$  (exponent  $n$ ), five distinct primes. Define the  $P_a$ – $P_b$  interaction norm

$$N_{u=0} := \min \left\{ \max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : \begin{array}{l} k_1\varphi_p + k_2\varphi_q + j_1\varphi_r + j_2\varphi_s = 0, \\ (\varphi_r + \varphi_s) \neq (\varphi_p + \varphi_q) \end{array} \right\}.$$

Let  $B_{\varphi_q=0}$  (resp.  $B_{\varphi_p=0}$ ) be the exact  $\text{nd}_{\text{val}}$  of the type-(1, 2, 1) sub-problem on primes  $(p, r, s, u)$  with exponents  $(k_1, j_1, j_2, n)$  (resp.  $(q, r, s, u)$  with  $(k_2, j_1, j_2, n)$ ), computed via Candidate Formula 3. Let  $B_{\varphi_s=0}$  (resp.  $B_{\varphi_r=0}$ ) be the exact  $\text{nd}_{\text{val}}$  of the type-(2, 1, 1) sub-problem on primes  $(p, q, r, u)$  with exponents  $(k_1, k_2, j_1, n)$  (resp.  $(p, q, s, u)$  with  $(k_1, k_2, j_2, n)$ ), computed via Candidate Formula 2. Then

$$\text{nd}_{\text{val}}(a, b) = \min(N_{u=0}, B_{\varphi_q=0}, B_{\varphi_p=0}, B_{\varphi_s=0}, B_{\varphi_r=0}).$$

*Evidence. Upper bound.* Each branch supplies a constructive non-degenerate witness:  $N_{u=0}$  by Bezout search on the four-prime homogeneous lattice;  $B_{\varphi_q=0}$ ,  $B_{\varphi_p=0}$  by Candidate Formula 3;  $B_{\varphi_s=0}$ ,  $B_{\varphi_r=0}$  by Candidate Formula 2.

*Lower bound.* Any non-degenerate  $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s, \varphi_u)$  with  $k_1\varphi_p + k_2\varphi_q + j_1\varphi_r + j_2\varphi_s = n\varphi_u$  and  $W = (\varphi_r + \varphi_s) - (\varphi_p + \varphi_q) \neq 0$  satisfies at least one of:

- $\varphi_u = 0$ : norm  $\geq N_{u=0}$  by definition.
- $\varphi_q = 0$ :  $\varphi$  lies in the type-(1, 2, 1) sub-lattice for  $(p^{k_1}, r^{j_1}, s^{j_2}, u^n)$ , so norm  $\geq B_{\varphi_q=0}$ .
- $\varphi_p = 0$ : analogously norm  $\geq B_{\varphi_p=0}$ .
- $\varphi_s = 0$ :  $\varphi$  lies in the type-(2, 1, 1) sub-lattice for  $(p^{k_1}, q^{k_2}, r^{j_1}, u^n)$ , so norm  $\geq B_{\varphi_s=0}$ .
- $\varphi_r = 0$ : analogously norm  $\geq B_{\varphi_r=0}$ .
- All five coordinates non-zero: verified computationally (script T79, 1425 triples in  $[4, 300) \times [4, 300)$ , spot-check 60, 0 failures).

□

**Remark 16.** The five branches correspond to zeroing each prime-group slot in turn.  $N_{u=0}$  subsumes the within- $P_a$ , within- $P_b$ , and all  $P_a$ – $P_b$  cross-group pairwise norms; the type-(1, 2, 1) and type-(2, 1, 1) sub-problem branches subsume all  $P_a$ – $P_c$  and  $P_b$ – $P_c$  interactions. Script T79 (1425 triples, 0 failures, spot-check against brute-force):  $N_{u=0}$  achieves the minimum in 1405 cases (98.6%);  $B_{\varphi_s=0}$  or  $B_{\varphi_r=0}$  wins in the remaining 20 cases.

**Candidate Formula 5** (Exact  $\text{nd}_{\text{val}}$  for type  $(1, 2, 2)$  triples). *Let  $a + b = c$  be coprime with  $P_a = \{p\}$  (exponent  $k$ ),  $P_b = \{q, r\}$  (exponents  $j, m$ ),  $P_c = \{s, t\}$  (exponents  $j_1, j_2$ ), five distinct primes. The Wronskian is  $W = (\varphi_q + \varphi_r) - \varphi_p$ . Define the  $P_b$ – $P_c$  interaction norm*

$$N_{p=0} := \min\{\max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : j\varphi_q + m\varphi_r - j_1\varphi_s - j_2\varphi_t = 0, \varphi_q + \varphi_r \neq 0\}.$$

*Let  $B_{\varphi_r=0}$  (resp.  $B_{\varphi_q=0}$ ) be the exact  $\text{nd}_{\text{val}}$  of the type- $(1, 1, 2)$  sub-problem on primes  $(p, q, s, t)$  with exponents  $(k, j, j_1, j_2)$  (resp.  $(p, r, s, t)$  with  $(k, m, j_1, j_2)$ ), via Candidate Formula 1. Let  $B_{\varphi_t=0}$  (resp.  $B_{\varphi_s=0}$ ) be the exact  $\text{nd}_{\text{val}}$  of the type- $(1, 2, 1)$  sub-problem on primes  $(p, q, r, s)$  with exponents  $(k, j, m, j_1)$  (resp.  $(p, q, r, t)$  with  $(k, j, m, j_2)$ ), via Candidate Formula 3. Then*

$$\text{nd}_{\text{val}}(a, b) = \min(N_{p=0}, B_{\varphi_r=0}, B_{\varphi_q=0}, B_{\varphi_t=0}, B_{\varphi_s=0}).$$

*Evidence. Upper bound.* Each branch supplies a constructive witness:  $N_{p=0}$  by Bezout search on the four-prime homogeneous lattice;  $B_{\varphi_r=0}, B_{\varphi_q=0}$  by Candidate Formula 1;  $B_{\varphi_t=0}, B_{\varphi_s=0}$  by Candidate Formula 3.

*Lower bound.* Any non-degenerate  $\varphi = (\varphi_p, \varphi_q, \varphi_r, \varphi_s, \varphi_t)$  with  $k\varphi_p + j\varphi_q + m\varphi_r = j_1\varphi_s + j_2\varphi_t$  and  $W = (\varphi_q + \varphi_r) - \varphi_p \neq 0$  satisfies at least one of:

- $\varphi_p = 0$ : norm  $\geq N_{p=0}$  by definition.
- $\varphi_r = 0$ :  $\varphi$  lies in the type- $(1, 1, 2)$  sub-lattice for  $(p^k, q^j, s^{j_1}, t^{j_2})$ , so norm  $\geq B_{\varphi_r=0}$ .
- $\varphi_q = 0$ : analogously norm  $\geq B_{\varphi_q=0}$ .
- $\varphi_t = 0$ :  $\varphi$  lies in the type- $(1, 2, 1)$  sub-lattice for  $(p^k, \{q^j, r^m\}, s^{j_1})$ , so norm  $\geq B_{\varphi_t=0}$ .
- $\varphi_s = 0$ : analogously norm  $\geq B_{\varphi_s=0}$ .
- All five coordinates non-zero: verified computationally (script T80, 996 triples in  $[2, 150) \times [4, 150)$ , spot-check 50, 0 failures).

□

**Remark 17.** *Script T80 (996 triples, 0 failures):  $N_{p=0}$  achieves the minimum in 869 cases (87.2%);  $B_{\varphi_r=0}$  wins in 72 cases (7.2%),  $B_{\varphi_q=0}$  in 52 (5.2%),  $B_{\varphi_t=0}$  in 3 (0.3%).*

**Candidate Formula 6** (Exact  $\text{nd}_{\text{val}}$  for type  $(2, 1, 2)$  triples). *Let  $a + b = c$  be coprime with  $P_a = \{p, q\}$  (exponents  $k_1, k_2$ ),  $P_b = \{r\}$  (exponent  $m$ ),  $P_c = \{s, t\}$  (exponents  $j_1, j_2$ ). By relabelling  $a \leftrightarrow b$  (so  $P_a \leftrightarrow P_b$ ), this is equivalent to a type- $(1, 2, 2)$  triple with the same five primes. Applying Candidate Formula 5 with the label swap gives*

$$\text{nd}_{\text{val}}(a, b) = \min(N_{r=0}, B_{\varphi_q=0}, B_{\varphi_p=0}, B_{\varphi_t=0}, B_{\varphi_s=0}),$$

*where  $N_{r=0}$  is the  $P_a$ – $P_c$  homogeneous interaction (zeroing  $\varphi_r$ ),  $B_{\varphi_q=0}$  and  $B_{\varphi_p=0}$  are type- $(1, 1, 2)$  sub-problems via Candidate Formula 1, and  $B_{\varphi_t=0}, B_{\varphi_s=0}$  are type- $(2, 1, 1)$  sub-problems via Candidate Formula 2.*

**Candidate Formula 7** (Exact  $\text{nd}_{\text{val}}$  for type  $(1, 1, 3)$  triples). Let  $a + b = c$  be coprime with  $P_a = \{p\}$  (exponent  $k$ ),  $P_b = \{q\}$  (exponent  $m$ ),  $P_c = \{r, s, t\}$  (exponents  $n_1, n_2, n_3$ ), five distinct primes. The constraint is  $k\varphi_p + m\varphi_q = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t$  and  $W = \varphi_q - \varphi_p$ . Define the one-sided interaction norms

$$\begin{aligned} N_{p=0} &:= \min\{\max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : m\varphi_q = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t, \varphi_q \neq 0\}, \\ N_{q=0} &:= \min\{\max(p|\varphi_p|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : k\varphi_p = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t, \varphi_p \neq 0\}. \end{aligned}$$

Let  $\text{nd}_4(\cdot)$  denote the type- $(1, 1, 2)$  formula (Candidate Formula 1). Then

$$\text{nd}_{\text{val}}(a, b) = \min\left(\text{nd}_4(p^k, q^m, s^{n_2}, t^{n_3}), \text{nd}_4(p^k, q^m, r^{n_1}, t^{n_3}), \text{nd}_4(p^k, q^m, r^{n_1}, s^{n_2}), N_{p=0}, N_{q=0}\right).$$

*Evidence. Upper bound.* The three  $\text{nd}_4$  branches use Candidate Formula 1 on the sub-triple obtained by setting one  $c$ -prime coordinate to zero;  $N_{p=0}$  takes  $\varphi_p = 0$  with  $W = \varphi_q \neq 0$ ;  $N_{q=0}$  takes  $\varphi_q = 0$  with  $W = -\varphi_p \neq 0$ .

*Lower bound.* Any non-degenerate  $\varphi$  satisfies at least one of:

- $\varphi_r = 0$ :  $\varphi$  lies in the type- $(1, 1, 2)$  sub-lattice for  $(p^k, q^m, s^{n_2}t^{n_3})$ , so  $\text{norm} \geq \text{nd}_4(p^k, q^m, s^{n_2}, t^{n_3})$ .
- $\varphi_s = 0$ : analogously  $\text{norm} \geq \text{nd}_4(p^k, q^m, r^{n_1}, t^{n_3})$ .
- $\varphi_t = 0$ : analogously  $\text{norm} \geq \text{nd}_4(p^k, q^m, r^{n_1}, s^{n_2})$ .
- $\varphi_p = 0$ :  $W = \varphi_q \neq 0$ , so  $m\varphi_q = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t$  with  $\varphi_q \neq 0$ , hence  $\text{norm} \geq N_{p=0}$ .
- $\varphi_q = 0$ :  $W = -\varphi_p \neq 0$ , so  $k\varphi_p = n_1\varphi_r + n_2\varphi_s + n_3\varphi_t$  with  $\varphi_p \neq 0$ , hence  $\text{norm} \geq N_{q=0}$ .
- All five coordinates non-zero: verified computationally (script T81, 729 triples with  $a, b \leq 300$ , spot-check 60, 0 failures).

□

**Remark 18.** Script T81 verifies Candidate Formula 7 on all 729 type- $(1, 1, 3)$  triples with  $a, b \leq 300$ , spot-check 60: 0 failures. Branch win counts:  $\text{nd}_4(\cdot, r^{n_1}, s^{n_2})$  (zeroing  $\varphi_t$ ) in 669 cases (91.8%);  $\text{nd}_4(\cdot, r^{n_1}, t^{n_3})$  in 32 (4.4%);  $N_{q=0}$  in 16 (2.2%);  $N_{p=0}$  in 12 (1.6%);  $\text{nd}_4(\cdot, s^{n_2}, t^{n_3})$  in 0 cases.

**Candidate Formula 8** (Exact  $\text{nd}_{\text{val}}$  for type  $(1, 3, 1)$  triples). Let  $a + b = c$  be coprime with  $P_a = \{p\}$  (exponent  $k$ ),  $P_b = \{q, r, s\}$  (exponents  $n_1, n_2, n_3$ ),  $P_c = \{t\}$  (exponent  $m$ ), five distinct primes. The Wronskian is  $W = (\varphi_q + \varphi_r + \varphi_s) - \varphi_p$ . Define the  $P_a$ - $P_b$  homogeneous norm

$$N_{t=0} := \min\left\{\max(p|\varphi_p|, q|\varphi_q|, r|\varphi_r|, s|\varphi_s|) : \begin{aligned} &k\varphi_p + n_1\varphi_q + n_2\varphi_r + n_3\varphi_s = 0, \\ &\varphi_q + \varphi_r + \varphi_s \neq \varphi_p \end{aligned} \right\}$$

and the  $P_b$ - $P_c$  interaction norm

$$N_{p=0} := \min \left\{ \max(q|\varphi_q|, r|\varphi_r|, s|\varphi_s|, t|\varphi_t|) : \begin{array}{l} n_1\varphi_q + n_2\varphi_r + n_3\varphi_s = m\varphi_t, \\ \varphi_q + \varphi_r + \varphi_s \neq 0 \end{array} \right\}.$$

Let  $B_{\varphi_q=0}$  (resp.  $B_{\varphi_r=0}$ ,  $B_{\varphi_s=0}$ ) be the exact  $\text{nd}_{\text{val}}$  of the type- $(1, 2, 1)$  sub-problem on primes  $(p, r, s, t)$  with exponents  $(k, n_2, n_3, m)$  (resp.  $(p, q, s, t)$  with  $(k, n_1, n_3, m)$ ;  $(p, q, r, t)$  with  $(k, n_1, n_2, m)$ ), computed via Candidate Formula 3. Then

$$\text{nd}_{\text{val}}(a, b) = \min(N_{t=0}, N_{p=0}, B_{\varphi_q=0}, B_{\varphi_r=0}, B_{\varphi_s=0}).$$

*Evidence. Upper bound.*  $N_{t=0}$  takes  $\varphi_t = 0$  with  $W \neq 0$ ;  $N_{p=0}$  takes  $\varphi_p = 0$  with  $W = \varphi_q + \varphi_r + \varphi_s \neq 0$ ; each  $B_{\varphi_\bullet=0}$  uses Candidate Formula 3 on the sub-lattice obtained by setting one  $b$ -prime coordinate to zero.

*Lower bound.* Any non-degenerate  $\varphi$  satisfies at least one of:

- $\varphi_t = 0$ : norm  $\geq N_{t=0}$  by definition.
- $\varphi_p = 0$ :  $W = \varphi_q + \varphi_r + \varphi_s \neq 0$ , so norm  $\geq N_{p=0}$ .
- $\varphi_q = 0$ :  $\varphi$  lies in the type- $(1, 2, 1)$  sub-lattice for  $(p^k, r^{n_2}, s^{n_3}, t^m)$ , so norm  $\geq B_{\varphi_q=0}$ .
- $\varphi_r = 0$ : analogously norm  $\geq B_{\varphi_r=0}$ .
- $\varphi_s = 0$ : analogously norm  $\geq B_{\varphi_s=0}$ .
- All five coordinates non-zero: verified computationally (script T82, 843 triples with  $a, b \leq 300$ , spot-check 60, 0 failures).

□

**Remark 19.** Script T82 verifies Candidate Formula 8 on all 843 type- $(1, 3, 1)$  triples with  $a, b \leq 300$ , spot-check 60: 0 failures. Branch win counts:  $N_{t=0}$  in 803 cases (95.3%);  $N_{p=0}$  in 40 (4.7%); the three  $B_{\varphi_\bullet=0}$  branches in 0 cases. The empirical two-branch formula  $\min(N_{t=0}, N_{p=0})$  holds on all 843 surveyed triples; independently confirmed by a direct spot-check (seed 91, 60 triples, 0 failures).

**Candidate Formula 9** (Exact  $\text{nd}_{\text{val}}$  for type  $(3, 1, 1)$  triples). Let  $a + b = c$  be coprime with  $P_a = \{p, q, r\}$  (exponents  $k_1, k_2, k_3$ ),  $P_b = \{s\}$  (exponent  $m$ ),  $P_c = \{t\}$  (exponent  $n$ ), five distinct primes. The exact  $\text{nd}_{\text{val}}$  formula is obtained from Candidate Formula 8 by the  $P_a \leftrightarrow P_b$  label swap:

$$\text{nd}_{\text{val}}(a, b) = \min(\tilde{N}_{t=0}, \tilde{N}_{s=0}, \tilde{B}_{\varphi_p=0}, \tilde{B}_{\varphi_q=0}, \tilde{B}_{\varphi_r=0}),$$

where  $\tilde{N}_{t=0}$  is the  $P_a$ - $P_b$  homogeneous norm (zero  $\varphi_t$ ,  $W \neq 0$ ),  $\tilde{N}_{s=0}$  is the  $P_a$ - $P_c$  interaction norm (zero  $\varphi_s$ ,  $\varphi_p + \varphi_q + \varphi_r \neq 0$ ), and  $\tilde{B}_{\varphi_p=0}$ ,  $\tilde{B}_{\varphi_q=0}$ ,  $\tilde{B}_{\varphi_r=0}$  are type- $(1, 2, 1)$  sub-problems via Candidate Formula 3.

*Evidence.* The constraint for type  $(3, 1, 1)$  is  $k_1\varphi_p + k_2\varphi_q + k_3\varphi_r + m\varphi_s = n\varphi_t$  with  $W = \varphi_s - (\varphi_p + \varphi_q + \varphi_r)$ . Under  $a \leftrightarrow b$  this maps to the type- $(1, 3, 1)$  setup of Candidate Formula 8 with  $P_a \leftrightarrow P_b$  and  $W \rightarrow -W$ ; non-degeneracy is symmetric in sign of  $W$ , so Candidate Formula 8 applies verbatim. □

## Zero-coordinate minimizer property

The results in this and the following subsection that are printed as *Candidate Formula* are finite-verification formulas: their displayed witnesses and branch reductions are proved, but the all-nonzero coordinate case has only been checked on the stated finite sample. They must not be used as unconditional theorems.

**Theorem 30** (Squarefree minimizer support). *Let  $a + b = c$  be a squarefree coprime triple ( $v_p(abc) = 1$  for all  $p$ ) with  $\omega^* \geq 3$  distinct primes. Let  $m_1 \leq m_2 \leq m_3$  be the sorted group minima  $\min(P_a), \min(P_b), \min(P_c)$ . Then:*

- (i)  $\text{nd}(a, b) \leq m_2$  (Corollary 16: cross-group pair  $(m_1, m_2)$  with  $\gcd = 1$  gives a non-degenerate vector of norm  $\max(m_1, m_2) = m_2$ ).
- (ii) Every nd-minimizer  $\varphi$  satisfies  $\varphi_p = 0$  for every prime  $p > m_2$ . (Proof:  $p|\varphi_p| \leq \text{nd} \leq m_2 < p$ , so  $|\varphi_p| < 1$ , forcing  $\varphi_p = 0$ .)
- (iii) At least  $\omega^* - 2$  coordinates of every minimizer are zero. In particular  $\text{nd}(a, b) = \min_p B_p$  (zero-coordinate property, proved for all squarefree triples of any  $\omega^* \geq 3$ ).

*Proof.* (i) is the squarefree case of Corollary 16 (all valuations 1,  $\gcd = 1$ ,  $\text{norm} = m_2$ ).

(ii) For any prime  $p > m_2$ : since  $m_1 \leq m_2 < m_3 \leq p$  and  $\text{nd} \leq m_2$ , we have  $p|\varphi_p| \leq \text{nd} \leq m_2 < p$ , giving  $|\varphi_p| < 1$ , hence  $\varphi_p = 0$ .

(iii) Among the  $\omega^*$  primes, at most two satisfy  $p \leq m_2$  (namely  $m_1$  and  $m_2$  themselves; all other group elements exceed  $m_2$  by minimality of group minima within their groups). Hence at least  $\omega^* - 2$  primes exceed  $m_2$  and are forced to zero. For  $\omega^* \geq 3$ :  $\omega^* - 2 \geq 1$ , so every minimizer has at least one zero coordinate, i.e.  $\text{nd}(a, b) \geq \min_p B_p$ . The reverse inequality  $\text{nd} \leq \min_p B_p$  is immediate (each  $B_p$  is a minimum over a sub-lattice).  $\square$

**Remark 20.** *The squarefree hypothesis is essential. For non-squarefree  $\omega^* = 3$  triples,  $\text{nd}$  can exceed all primes in the triple (e.g.  $(5, 27, 32)$ : primes  $\{2, 3, 5\}$ ,  $\text{nd} = 6 > 5$ ), so the integrality argument fails and all-nonzero minimizers can occur.*

The individual exact nd formulas (Candidate Formulas 2–9) share a common structural feature: in every verified case, the minimum non-degenerate norm is achieved by a lattice vector with at least one prime coordinate equal to zero. We record the non-squarefree instances as a finite-verification candidate.

**Candidate Formula 10** (Zero-coordinate minimizer property). *Let  $a + b = c$  be a coprime triple with  $\omega^*$  distinct primes. Define, for each prime  $p \in P_a \cup P_b \cup P_c$ ,*

$$B_p := \min_q \{ \max_q q|\varphi_q| : \varphi \in F_{\text{val}}(a, b), \varphi_p = 0, W(\varphi) \neq 0 \}.$$

*In parts (ii)–(iv),  $F(a, b)$  is the valuation-coordinate lattice  $F_{\text{val}}(a, b)$  and  $\text{nd}$  means  $\text{nd}_{\text{val}}$ . Then:*

- (i) For squarefree triples of any  $\omega^* \geq 3$ :  $\text{nd}(a, b) = \min_p B_p$  (proved, Theorem 30).
- (ii) For non-squarefree  $\omega^* = 4$ :  $\text{nd}_{\text{val}}(a, b) = \min_p B_p$  (supported by Candidate Formulas 2, 3, and 1, each with the all-nonzero case verified only on the stated finite sample).

(iii) For non-squarefree  $\omega^* \in \{5, 6\}$ :  $\text{nd}_{\text{val}}(a, b) = \min_p B_p$  on the finite samples described below. Both cases use the  $p_k$ -forcing argument (primes labelled  $p_1 < \dots < p_k$ ,  $k = \omega^*$ ): any all-nonzero  $\varphi$  with  $\|\varphi\| < B_{p_k}$  satisfies  $p_k |\varphi_{p_k}| \leq \|\varphi\| < p_k$ , forcing  $\varphi_{p_k} = 0$ . Thus it suffices to verify  $B_{p_k} \leq p_k$ .

$\omega^* = 5$  (2446 non-squarefree triples,  $a, b \leq 150$ ): 2019 (82.5%) via a squarefree cross-group pair in  $\{p_1, \dots, p_4\}$  (analytic, T86bis sub-case B1); 427 (17.5%) via  $B_{p_5} \leq p_5$  directly (T86bis,  $BB = 8$ , 0 violations, 0 skips).

$\omega^* = 6$  (1959 non-squarefree triples,  $a, b \leq 200$ ): 1920 (98.0%) via a squarefree cross-group pair in  $\{p_1, \dots, p_5\}$  (analytic, T85bis sub-case B1); 39 (2.0%) via  $B_{p_6} \leq p_6$  directly (T85bis,  $BB = 8$ , 0 violations, 0 skips).

(iv) For non-squarefree  $\omega^* = 3$ : the identity fails. The triple  $(5, 27, 32)$  has  $\text{nd}_{\text{val}} = 6$  (achieved by  $\varphi = (-1, 2, 1)$ , all coordinates non-zero), while  $\min_p B_p = 15$ .

*Evidence. Part (i).* Theorem 30: integrality forces  $\varphi_p = 0$  for all  $p > \text{nd} \leq m_2$  (second group minimum).

*Part (ii).* For type  $(2, 1, 1)$ : Candidate Formula 2 decomposes  $\text{nd}$  into four branches each zeroing one prime ( $N_{s=0}$ ,  $N_{r=0}$ , two  $\text{nd}$ -111 sub-problems with  $\varphi_q = 0$  or  $\varphi_p = 0$ ). Types  $(1, 2, 1)$  and  $(1, 1, 2)$  follow analogously from Candidate Formulas 3 and 1.

*Part (iii).* Label the primes  $p_1 < \dots < p_k$  ( $k = \omega^* \in \{5, 6\}$ ). The  $p_k$ -forcing argument applies uniformly: any all-nonzero  $\varphi \in F_{\text{val}}(a, b)$  with  $W(\varphi) \neq 0$  satisfies  $p_k |\varphi_{p_k}| \leq \|\varphi\|$ ; if  $\|\varphi\| < B_{p_k}$  then  $p_k |\varphi_{p_k}| < p_k$ , forcing  $\varphi_{p_k} = 0$  — contradiction. Hence  $\text{nd}(a, b) \geq B_{p_k}$ , and equality  $\text{nd}(a, b) = \min_p B_p$  holds as soon as  $B_{p_k} \leq p_k$ .

*Sub-case B1* (analytic, 82.5% for  $\omega^* = 5$ ; 98.0% for  $\omega^* = 6$ ): there exist  $p_i, p_j \in \{p_1, \dots, p_{k-1}\}$  with  $v_{p_i}(abc) = v_{p_j}(abc) = 1$  lying in different groups. The cross-group Bézout vector  $(\varphi_{p_i}, \varphi_{p_j})$  the unique non-zero coordinates, all others 0) has  $\varphi_{p_k} = 0$ ,  $W \neq 0$ , and norm  $\max(p_i, p_j) \leq p_{k-1} < p_k$ . Hence  $B_{p_k} \leq p_{k-1} < p_k$ .

*Sub-case B2* ( $\omega^* = 5$ : 427 triples;  $\omega^* = 6$ : 39 triples): no squarefree cross-group pair exists in  $\{p_1, \dots, p_{k-1}\}$ . Brute-force search on the zero- $p_k$  sub-lattice with  $\|\varphi\|_\infty \leq 8$  verifies  $B_{p_k} \leq p_k$  for all such triples (T86bis  $\omega^* = 5$ ; T85bis  $\omega^* = 6$ ; 0 violations, 0 skips).

*Part (iv).* For  $(a, b, c) = (5, 27, 32)$ :  $P_a = \{5\}$ ,  $P_b = \{3\}$ ,  $P_c = \{2\}$ , exponents 1, 3, 5. The vector  $\varphi = (-1, 2, 1)$  satisfies the lattice constraint  $(1 \cdot (-1) + 3 \cdot 2 = 5 \cdot 1)$ ,  $W = 2 - (-1) = 3 \neq 0$ , and  $\|\varphi\| = \max(5, 6, 2) = 6$ . The three zero-coordinate branches give norms 15, 25, 15 (computed from the one-prime-per-group base case), so  $\min_p B_p = 15 > 6$ .  $\square$

**Remark 21.** The failure at  $\omega^* = 3$  is explained by the pairwise-Bézout regime (Theorem 28): when  $p_L$  is the largest prime and  $\gcd(\text{other exponents}) \mid k_{p_L}$ , the minimizer has  $|\varphi_{p_L}| = 1$  and all other coordinates nonzero. For  $\omega^* \geq 4$  this all-nonzero configuration at the base level is subsumed: the zero- $\varphi_{p_L}$  branch reduces to a  $(\omega^* - 1)$ -prime sub-problem whose own Bézout regime provides a vector at least as good.

## General recursive $\text{nd}$ algorithm

Candidate Formula 10 and the individual exact- $\text{nd}$  results suggest the following recursion for  $\text{nd}(a, b)$ ; its non-squarefree branches concern  $\text{nd}_{\text{val}}$  and are finite-verification claims outside the squarefree case.

**Candidate Formula 11** (General recursive nd formula). *Let  $a + b = c$  be a coprime triple with  $\omega^* \geq 3$  distinct primes partitioned as  $P_a \cup P_b \cup P_c$ . For each prime  $p \in P_a \cup P_b \cup P_c$ , define the  $p$ -branch  $B_p$  recursively:*

- (Multi-prime reduction) *If  $|P_x| \geq 2$  for the group  $P_x$  containing  $p$ :  $B_p = \text{nd}(a', b')$  where  $(a', b', c')$  is the  $(\omega^* - 1)$ -prime sub-problem obtained by removing prime  $p$  from  $P_x$  (its coordinate is forced to zero).*
- (Singleton interaction) *If  $|P_x| = 1$ :  $B_p$  is the group-interaction minimum  $\min\{\max_q q|\varphi_q| : \varphi \in F_{\text{val}}(a, b), \varphi_p = 0, W(\varphi) \neq 0\}$ , a direct Bezout-type minimization over the remaining two-group constraint.*

The base case ( $\omega^* = 3$ ) is Theorem 28. Then:

- (i) *For non-squarefree  $\omega^* = 4$ :  $\text{nd}_{\text{val}}(a, b) = \min_p B_p$  (supported by Candidate Formulas 2, 3, 1).*
- (ii) *For non-squarefree  $\omega^* \in \{5, 6\}$ :  $\text{nd}_{\text{val}}(a, b) = \min_p B_p$  (empirically verified for non-squarefree;  $\omega^* = 5$  both partition types  $(2, 2, 1)$  and  $(3, 1, 1)$ ,  $a, b \leq 150$ , T86bis;  $\omega^* = 6$  all three unordered partition classes  $\{4, 1, 1\}$ ,  $\{3, 2, 1\}$ ,  $\{2, 2, 2\}$ : 1959 non-squarefree triples  $a, b \leq 200$ , 0 violations, T85bis; squarefree  $\omega^* \in \{5, 6\}$ : proved, Theorem 30).*

*Evidence.* Upper bound ( $\text{nd} \leq \min_p B_p$ ). Each  $B_p$  is a minimum over a sub-lattice of  $F(a, b)$ , hence  $\text{nd}(a, b) \leq B_p$  for every  $p$ .

*Lower bound for non-squarefree  $\omega^* = 4$  ( $\text{nd}_{\text{val}} \geq \min_p B_p$ ).* This is supported by Candidate Formula 10(ii): in every surveyed minimizer some coordinate is zero, placing it in one of the  $\omega^* = 4$  branches. The exact-nd sub-problem formulas (Candidate Formulas 1–2) confirm the identity.

*Part (ii).* Computational verification; see Candidate Formula 10(ii) for the zero-coordinate property and the individual T-scripts for the exact values.  $\square$

For  $\omega = 3$  type  $(1, 1, 1)$  we can characterize not just the minimum but also the *second* minimum non-degenerate norm.

**Theorem 31** (F32a: second minimum for type  $(1, 1, 1)$ ). *For a type  $(1, 1, 1)$  triple with primes  $p < q < r = p + q$ , the second minimum non-degenerate norm is  $r = q + 2$  (achieved by the vector  $\psi = (0, q, r)$ , i.e.  $\varphi = (0, 1, 1)$ , which has  $W = pq \neq 0$  and  $\|\psi\|_\infty = r$ ). Moreover there is no non-degenerate vector with norm strictly between  $q$  and  $r$ .*

*Proof. Upper bound.* The vector  $\varphi = (0, 1, 1)$  satisfies the lattice constraint  $S_a + S_b = S_c$  (i.e.  $0 + 1 = 1$ ) and is non-degenerate since  $S_b = 1 \neq 0 = S_a$ . Its norm is  $\max(0, q, r) = r = q + 2$ .

*Lower bound (no gap).* Suppose  $\psi$  is non-degenerate with  $\|\psi\|_\infty \in (q, r)$ . Since  $r|\varphi_r|$  must be strictly less than  $r$ , we have  $\varphi_r = 0$ . The constraint then gives  $\varphi_q = -\varphi_p$ . Non-degeneracy ( $S_b \neq S_a$ ) forces  $\varphi_p \neq 0$ . But the norm is  $\max(p|\varphi_p|, q|\varphi_p|, 0) = q|\varphi_p|$ . For the norm to lie strictly in  $(q, r) = (q, q + 2)$  we need  $q < q|\varphi_p| < q + 2$ , i.e.  $1 < |\varphi_p| < 1 + 2/q$ . Since  $q \geq 3$  (the smallest odd prime gap gives  $q = 3$ ),  $2/q \leq 2/3 < 1$ , so there is no integer in  $(1, 1 + 2/q)$ . Contradiction.

The integer core ( $q + 2 \leq kq$  for all  $k \geq 2$  and  $q \geq 3$ , with no integer  $k$  satisfying  $q < kq < q + 2$ ) is formalized as `pasten_F32a_gap` and `pasten_F32a_upper` in the Lean kernel.  $\square$

**Theorem 32** (F32b: complete norm spectrum for type  $(1, 1, 1)$ ,  $q \geq 5$ ). *Let  $(2, q, r)$  be a twin-prime triple with  $q \geq 5$ . The set of achievable non-degenerate norms is exactly*

$$\{kq : k \geq 1\} \cup \{kr : k \geq 1\}.$$

*Proof. Achievability.* The vector  $\varphi = (k, -k, 0)$  has norm  $kq$  (non-degenerate since  $S_b = -k \neq k = S_a$ );  $\varphi = (0, k, k)$  has norm  $kr$ .

*Completeness.* Let  $(\alpha, \beta) \in \mathbb{Z}^2$  with  $\alpha \neq \beta$ ,  $\gamma := \alpha + \beta$ . If  $\gamma = 0$  then norm  $= q|\alpha|$ , done. Suppose  $\gamma \neq 0$ . We show  $\max(q|\beta|, r|\gamma|) \geq p|\alpha| = 2|\alpha|$ ; this forces the norm to equal  $q|\beta|$  or  $r|\gamma|$ , hence a multiple of  $q$  or  $r$ .

Assume for contradiction that  $2|\alpha| > r|\gamma|$  and  $2|\alpha| > q|\beta|$ . By the triangle inequality  $|\beta| \geq |\alpha| - |\gamma|$ , so  $q|\beta| \geq q(|\alpha| - |\gamma|)$ . The second assumption gives  $q(|\alpha| - |\gamma|) < 2|\alpha|$ , i.e.  $(q - 2)|\alpha| < q|\gamma|$ . The first assumption gives  $|\gamma| < 2|\alpha|/r = 2|\alpha|/(q + 2)$ . Substituting:

$$(q - 2)|\alpha| < q \cdot \frac{2|\alpha|}{q+2} \implies (q - 2)(q + 2) < 2q \implies q^2 - 4 < 2q \implies q^2 - 2q - 4 < 0.$$

For  $q \geq 5$  this contradicts  $q^2 \geq 2q + 5$  (formalized as `pasten.F32b_ineq` in the Lean kernel).

*Exception.* The triple  $(2, 3, 5)$  ( $q = 3 < 2p = 4$ ) is the unique exception:  $\varphi = (8, -5, 3)$  gives norm 16, not a multiple of 3 or 5. For all other twin-prime triples ( $q \geq 5$ ), the spectrum is exactly  $\{kq\} \cup \{kr\}$  (verified numerically for  $q \leq 73$ , `t54`).  $\square$

**Corollary 33** (F32c: no Frobenius number for the non-degenerate spectrum). *For a twin-prime triple  $(2, q, r = q + 2)$  with  $q \geq 5$ , the set of positive integers that are not non-degenerate norms is infinite. In particular, the non-degenerate norm spectrum has no Frobenius number.*

*Proof.* By Theorem 32, the non-degenerate norms are exactly  $\{kq : k \geq 1\} \cup \{kr : k \geq 1\}$ . For example,  $q + r = 2q + 2$  is not divisible by either  $q$  or  $q + 2$  (for  $q \geq 5$ ), and the same is true for  $m(q + r)$  whenever  $\gcd(m, qr) = 1$ . Thus infinitely many positive integers are absent from the spectrum. The Sylvester–Frobenius number  $qr - q - r$  applies to the full numerical semigroup generated by  $q$  and  $r$ , not to this union of two rays; see also Remark 22.  $\square$

**Remark 22** (Semigroup structure). *The achievable non-degenerate norms  $\{kq\} \cup \{kr\}$  form a proper subset of the numerical semigroup  $\langle q, r \rangle$ . Elements of  $\langle q, r \rangle \setminus (\{kq\} \cup \{kr\})$ , such as  $q + r = 2q + 2$ , are achieved only by degenerate vectors: the vector  $\varphi = (k, k, 2k)$  satisfies  $S_b = k = S_a$ , giving  $W = 0$ . Theorem 32 shows that non-degeneracy exactly eliminates these mixed-term combinations, leaving only the pure scalings of  $q$  and  $r$ .*

**Remark 23** (Note on all type  $(1, 1, 1)$  triples having  $p = 2$ ). *For any type  $(1, 1, 1)$  triple  $(p, q, r)$  with  $p < q$ , we have  $r = p + q$ . If  $p \geq 3$ , both  $p$  and  $q$  are odd, so  $r = p + q$  is even and composite. Therefore every type  $(1, 1, 1)$  triple satisfies  $p = 2$ ; the twin-prime condition  $q$  and  $q + 2$  both prime applies universally.*

**F33: second minimum non-degenerate norm for  $\omega = 4$  type  $(1, 1, 2)$**

**Theorem 34** (F33: second minimum for type  $(1, 1, 2)$ ). *For type  $(1, 1, 2)$  with  $a = p$ ,  $b = q$ ,  $c = r_1 r_2$  ( $p < r_1 < r_2$  primes,  $p + q = r_1 r_2$ , minimum non-degenerate norm  $\text{nd} = r_1$ ):*

$$\text{second minimum non-degenerate norm} = \min(r_2, q, 2r_1).$$



*Proof.* Let  $M = \min(r_2, q, 2r_1)$ . We exhibit a vector of norm  $M$  and show no non-degenerate norm lies in  $(r_1, M)$ .

*Achievability.* If  $M = r_2$ :  $\varphi = (1, 0, 0, 1)$  satisfies the constraint ( $S_a + S_b = 1 = S_c$ ), is non-degenerate ( $S_b = 0 \neq 1 = S_a$ ), with norm  $\max(p, 0, 0, r_2) = r_2$ .

If  $M = q$ :  $\varphi = (1, -1, 0, 0)$  is non-degenerate with norm  $q$ .

If  $M = 2r_1$ :  $\varphi = (-2, 0, -2, 0)$  is non-degenerate with norm  $2r_1$ .

*Gap lemma.* Suppose  $\psi$  is non-degenerate with  $r_1 < \|\psi\|_\infty < M$ . Since  $r_2 \geq M$ , we have  $r_2|\varphi_{r_2}| < r_2$ , so  $\varphi_{r_2} = 0$ . Since  $q \geq M$ , we have  $q|\varphi_q| < q$ , so  $\varphi_q = 0$ . Non-degeneracy ( $\varphi_q \neq \varphi_p$ ) then forces  $\varphi_p \neq 0$ . The constraint gives  $\varphi_p = \varphi_{r_1}$ , so the norm equals  $r_1|\varphi_{r_1}| = r_1|\varphi_p|$ . For the norm to lie strictly in  $(r_1, M)$  we need  $r_1 < r_1k < M$  for some integer  $k = |\varphi_{r_1}| \geq 2$ . But  $k \geq 2$  gives  $r_1k \geq 2r_1 \geq M$  (since  $M \leq 2r_1$  by definition). Contradiction.

The integer core ( $M \leq kr_1$  for  $k \geq 2$  and  $M \leq 2r_1$ ) is formalized as `pasten_F33_gap` in the Lean kernel (zero `sorry`).  $\square$

**Theorem 35** (F36: Universal second minimum for squarefree triples). *Let  $a + b = c$  be a squarefree coprime triple with  $\omega^* \geq 3$  distinct prime divisors  $p_1 < p_2 < p_3 \leq \dots \leq p_{\omega^*}$ . Then  $\text{nd}(a, b) = p_2$  (Theorem 14) and*

$$\lambda_2(a, b) = \min(2p_2, p_3).$$

*In particular: for  $\omega^* = 3$  one always has  $p_3 = p_1 + p_2 < 2p_2$ , giving  $\lambda_2 = p_3$  (recovering Theorem 31); for  $\omega^* \geq 4$ , either  $2p_2 \leq p_3$  (gap formula  $\lambda_2 = 2p_2$ ) or  $p_3 < 2p_2$  (third-prime formula  $\lambda_2 = p_3$ ).*

*Proof. Lower bound.* Suppose  $\varphi$  is a non-degenerate vector with  $p_2 < \|\varphi\| < \min(2p_2, p_3)$ . Since  $p \cdot |\varphi_p| \leq \|\varphi\| < p_3 \leq p$  for every prime  $p \geq p_3$ , we have  $|\varphi_p| < 1$ , hence  $\varphi_p = 0$  for all  $p \geq p_3$ . The constraint  $\sum_i \alpha_i \varphi_i = 0$  then reduces to the two-variable equation  $\alpha_{p_1} \varphi_{p_1} + \alpha_{p_2} \varphi_{p_2} = 0$  with  $\alpha_{p_i} \in \{+1, -1\}$  (squarefree). The primitive solution is  $\varphi_{p_1} = \pm 1$ ,  $\varphi_{p_2} = \mp \alpha_{p_1} \alpha_{p_2}$ , achieving norm  $\max(p_1, p_2) = p_2$  (non-degeneracy holds since  $p_1$  and  $p_2$  lie in different prime groups when  $\text{nd} = p_2$ ). Every solution is an integer multiple of this primitive vector; hence every achievable norm on this branch is  $kp_2$  for  $k \in \mathbb{Z}_{>0}$ . No integer  $k$  satisfies  $p_2 < kp_2 < 2p_2$ , so no norm lies strictly in  $(p_2, 2p_2)$ . Together with  $\|\varphi\| < p_3$ , there is no norm in  $(p_2, \min(2p_2, p_3))$ .

*Upper bound. Achieving  $2p_2$ :* the  $\text{nd}$ -minimiser  $\varphi^*$  (with  $\|\varphi^*\| = p_2$  and  $W(\varphi^*) \neq 0$ ) satisfies  $W(2\varphi^*) = 2W(\varphi^*) \neq 0$  and  $\|2\varphi^*\| = 2p_2$ , so  $2p_2$  is an achievable non-degenerate norm.

*Achieving  $p_3$ :* by the squarefree zero-coordinate property (Theorem 30), the  $\text{nd}$ -minimising sub-problem for the branch  $\varphi_{p_1} = 0$  is a squarefree  $(\omega^* - 1)$ -prime sub-triple whose primes are  $\{p_2, \dots, p_{\omega^*}\}$ . By Theorem 14 applied to this sub-triple, its minimum non-degenerate norm equals  $p_3$  (the second-smallest element of  $\{p_2, \dots, p_{\omega^*}\}$ ), achieved by a vector  $\varphi'$  with  $\varphi'_{p_1} = 0$ ,  $\|\varphi'\| = p_3$ ,  $W(\varphi') \neq 0$ . Hence  $p_3$  is achievable.

Since both  $2p_2$  and  $p_3$  exceed  $\text{nd}(a, b) = p_2$ , we get  $\lambda_2 \leq \min(2p_2, p_3)$ , completing the proof.  $\square$

**Remark 24.** *Theorem 35 unifies F32a (Theorem 31), and Theorem 34 follows as a corollary: for type  $(1, 1, 2)$  with  $p < r_1 < r_2$  and  $q = r_1 r_2 - p \gg r_2$ , the sorted primes are  $p_1 = p, p_2 =$*

$r_1, p_3 = r_2, p_4 = q$  and Theorem 35 gives  $\lambda_2 = \min(2r_1, r_2)$ ; the “ $q$ ” branch in Theorem 34 is always dominated by  $r_2 \leq q$ . T90b verifies Theorem 35 on 778 squarefree  $\omega^* = 4$  triples with  $a, b \leq 400$  (zero failures), plus 42 triples at  $\omega^* = 5$  and 12 at  $\omega^* = 6$  (all zero failures).

**Theorem 36** (F37: Third successive minimum for squarefree type  $(1, 1, 2)$  triples). *Let  $a + b = c$  be a squarefree coprime triple of type  $(1, 1, 2)$  (i.e.  $|P_a| = |P_b| = 1$ ,  $|P_c| = 2$ ) with  $\omega^* = 4$  distinct prime divisors  $p_1 < p_2 < p_3 < p_4$ . Then  $\lambda_1 = p_2$  ( $E_n$  theorem),  $\lambda_2 = \min(2p_2, p_3)$  (Theorem 35), and*

$$\lambda_3(a, b) = \begin{cases} \min(2p_2, p_4) & \text{if } p_3 < 2p_2 \quad (\text{Case 1: } \lambda_2 = p_3), \\ \min(3p_2, p_3) & \text{if } p_3 \geq 2p_2 \quad (\text{Case 2: } \lambda_2 = 2p_2). \end{cases}$$

*Proof.* Write  $\lambda := \lambda_3(a, b)$ . We prove matching lower and upper bounds in each case.

*Upper bounds.* In both cases  $3p_2$  is achieved by tripling the nd-minimiser  $\varphi^*$  (non-degeneracy scales linearly). In Case 2,  $p_3$  is achieved by the sub-problem with  $\varphi_{p_1} = 0$ : the  $E_n$  theorem applied to  $\{p_2, p_3, p_4\}$  gives  $\text{nd} = p_3$ . In Case 1,  $2p_2$  is achieved by doubling  $\varphi^*$ . The bound  $p_4$  is achieved by an explicit non-degenerate vector with  $|\varphi_{p_4}| = 1$ : by the UD constraint  $\varphi_{p_a} + \varphi_{p_b} = \varphi_{p_{c,1}} + \varphi_{p_{c,2}}$  (with  $|P_a| = |P_b| = 1$ ,  $|P_c| = 2$ ), setting the coordinate in the same UD-pair as  $p_4$  to  $+1$  and all other coordinates to 0 satisfies UD and gives  $\varphi_{p_a} \neq \varphi_{p_b}$ ; the norm equals  $p_4$ . Hence  $\lambda \leq \min$  of the stated formula.

*Lower bounds. Case 1* ( $p_3 < 2p_2$ ,  $\lambda_2 = p_3$ ). We show no non-degenerate norm lies in  $(p_3, \min(2p_2, p_4))$ . Let  $N$  be such a norm with  $p_3 < N < \min(2p_2, p_4)$ . Since  $p_4 \cdot |\varphi_{p_4}| \leq N < p_4$ , we get  $\varphi_{p_4} = 0$ . Since  $p_3 \cdot |\varphi_{p_3}| \leq N < 2p_2$  and  $|\varphi_{p_3}| \geq 2$  would give  $p_3 \cdot |\varphi_{p_3}| \geq 2p_3 > 2p_2 > N$ , we get  $|\varphi_{p_3}| \leq 1$ . If  $|\varphi_{p_3}| = 1$ : any coordinate with  $p_2|\varphi_{p_2}| > p_3$  requires  $|\varphi_{p_2}| \geq 2$ , giving  $p_2|\varphi_{p_2}| \geq 2p_2 > N$  — contradiction. So  $|\varphi_{p_2}| \leq 1$  in either sub-case ( $\varphi_{p_3} = 0$  or  $|\varphi_{p_3}| = 1$ ). The UD constraint with  $\varphi_{p_4} = 0$ ,  $|\varphi_{p_3}| \leq 1$ ,  $|\varphi_{p_2}| \leq 1$  forces  $|\varphi_{p_1}| \leq 2$  (each group contributes at most 1 in absolute value to the UD sum), so  $p_1|\varphi_{p_1}| \leq 2p_1 < p_3 < N$  (three distinct primes satisfy  $2p_1 < p_3$ ); thus  $p_1$  does not determine the norm. The norm equals  $p_2|\varphi_{p_2}|$ . For  $N \in (p_3, 2p_2)$  this requires  $p_3/p_2 < |\varphi_{p_2}| < 2$ ; since  $p_3 > p_2$  gives  $p_3/p_2 > 1$ , no integer value of  $|\varphi_{p_2}|$  exists. Contradiction; so  $N$  does not exist and  $\lambda \geq \min(2p_2, p_4)$ .

*Case 2* ( $p_3 \geq 2p_2$ ,  $\lambda_2 = 2p_2$ ). We show no norm lies in  $(2p_2, \min(3p_2, p_3))$ . Let  $N$  be such a norm with  $2p_2 < N < \min(3p_2, p_3)$ . Since  $N < p_3 \leq p_i$  for all  $i \geq 3$ , we have  $\varphi_{p_3} = \varphi_{p_4} = 0$ . The norm reduces to the rank-2 sub-lattice  $\{p_1, p_2\}$  with achievable non-degenerate norms  $\{kp_2\}$ ; no integer  $k$  satisfies  $2p_2 < kp_2 < 3p_2$ . Contradiction; so  $\lambda \geq \min(3p_2, p_3)$ .  $\square$

**Remark 25.** T94b verifies Theorem 36 on all 360 squarefree type- $(1, 1, 2)$  coprime triples with  $c \leq 500$  (zero failures): 125 in Case 1 (including 71 where  $p_4 < 2p_2$  so the  $p_4$ -branch determines  $\lambda_3$ ) and 235 in Case 2. The two cases are exhaustive and mutually exclusive; Case 1 reduces to  $\lambda_3 = \min(2p_2, p_4)$  and Case 2 to  $\lambda_3 = \min(3p_2, p_3)$ , showing that  $\lambda_3$  is always determined by the “losing” branch of the  $\lambda_2$  competition together with the next available prime.

#### F11: a conditional unboundedness criterion at $\omega \geq 6$

**Theorem 37** (F11: conditional unboundedness criterion). *Let  $(n_a, n_b, n_c)$  be a partition type with  $n_a + n_b + n_c = \omega \geq 6$ . Suppose there is an infinite sequence of squarefree coprime triples*

of this type in which one constituent has at least three bounded prime factors and every prime outside that constituent tends to infinity while remaining within a fixed constant factor of the largest outside prime. Then

$$\rho = \frac{\|\psi_{\text{nd}}\|_{\infty}}{R^{1/(\omega-1)}} \rightarrow \infty.$$

In particular, the corresponding partition type is unbounded along that sequence.

*Proof.* By Theorem 11,  $\|\psi_{\text{nd}}\|_{\infty}$  is the second smallest of  $\{\min P_a, \min P_b, \min P_c\}$ . Under the stated hypothesis, every finite group minimum other than the bounded constituent's minimum tends to infinity; therefore the second-smallest finite group minimum is comparable to the outside scale  $N$ , and  $\|\psi_{\text{nd}}\|_{\infty} \asymp N$ . Since at least three primes remain bounded,  $R = O(N^{\omega-3})$ . Hence

$$\rho \gg \frac{N}{N^{(\omega-3)/(\omega-1)}} \rightarrow \infty. \quad \square$$

**Remark 26.** For every  $\omega \geq 6$  partition type, the required prime-pattern family is natural combinatorially and was checked on finite samples (for example, **t23** at  $\omega = 6$ ), but its infinitude is not proved here. Thus the criterion is conditional and must not be read as an unconditional classification of every  $\omega \geq 6$  type.

## F12–F16: upper bounds for bounded types

Combining the F10 formula with analytic optimization yields the following unconditional upper bounds for all universally-bounded types. Extremal constants that require an infinite prime-pattern family are conditional.

**Theorem 38** (Upper bounds for bounded types). *Let  $\rho = \|\psi_{\text{nd}}\|_{\infty}/R^{1/(\omega-1)}$ . For every squarefree coprime triple of each universally-bounded type,  $\rho$  is bounded as follows:*

| $\omega$ | Type      | upper bound              | equality in bound? | source |
|----------|-----------|--------------------------|--------------------|--------|
| 3        | (1, 1, 1) | $2^{-1/2} \approx 0.707$ | No                 | F16    |
| 4        | (0, 2, 2) | 1                        | No                 | F14    |
| 4        | (1, 1, 2) | $2^{-1/3} \approx 0.794$ | No                 | F14    |
| 4        | (1, 2, 1) | $2^{-1/3} \approx 0.794$ | No                 | F12    |
| 5        | (0, 2, 3) | 1                        | No                 | F23    |
| 5        | (0, 3, 2) | 1                        | No                 | F23    |
| 5        | (1, 2, 2) | $2^{-1/4} \approx 0.841$ | No                 | F26    |
| 5        | (2, 1, 2) | $2^{-1/4} \approx 0.841$ | No                 | F30    |
| 5        | (2, 2, 1) | $6^{-1/4} \approx 0.639$ | No                 | F24    |

In particular, every bounded type with  $a \geq 2$  has  $\rho^{\omega-1} < 1/2$ .

*Proof.* The (0, 2, 3), (0, 3, 2), (1, 2, 2), and (2, 2, 1) rows are Theorems 10, 8, and 9; the (2, 1, 2) row follows from the  $a \leftrightarrow b$  symmetry used in F30. The type-(0, 2, 2) bound follows by applying the F10 formula to its two non-empty group minima. For the balanced rows, let  $g \geq 2$  and  $d$  be the two smallest group minima. Every one of the  $\omega - 2$  remaining primes

is strictly larger than  $d$ , so  $R > 2d^{\omega-1}$  and  $\rho^{\omega-1} < 1/2$ . For completeness, we spell out the basic type-(1, 1, 1) estimate. For a prime triple  $(p, q, r)$  with  $p \leq q \leq r$  and  $p + q = r$ , parity forces  $p = 2$ . Then  $\rho^2 = q/(p(p+q)) = q/(2r)$ . Since  $q < r$ :  $\rho^2 < 1/2$ . This is Lean-formalized (`pasten_F16_111_ratio_sq_lt_half`). Conditional sharpness would follow from infinitely many twin pairs  $(q, q+2)$ ; their infinitude is open.  $\square$

*Proof sketch of the type-(0, 2, 2) row.* If the two-prime constituents are  $pq$  and  $rs$  with one even prime, then  $nd$  is the minimum of the odd constituent. The other three prime factors include a prime larger than  $nd$  in its own constituent and the two primes of the even constituent, whose product exceeds  $2nd$ ; hence  $R > nd^3$  and  $\rho < 1$ .  $\square$

## Unifying pattern

The balanced types follow a single upper-bound formula.

**Theorem 39** (Balanced-type universal upper bound, F16/F28). *For any partition type  $(1, k_1, k_2)$  with  $k_1 + k_2 = \omega - 1$  and any  $\omega \geq 3$ :*

$$\rho^{\omega-1} < \frac{1}{2},$$

*equivalently  $\rho < 2^{-1/(\omega-1)}$ . The bound is never attained.*

*Proof.* Let  $g$  be the smallest of the three group minima and let  $d = nd$  be the second-smallest group minimum. Then  $g \geq 2$ , and the  $\omega - 2$  primes other than  $g$  and  $d$  are all strictly greater than  $d$ :  $d$  is minimal in its group, while the third group minimum is at least  $d$  and distinct primes are unequal. Hence

$$\rho^{\omega-1} = \frac{d^{\omega-1}}{R} < \frac{d^{\omega-1}}{2 \cdot d \cdot d^{\omega-2}} = \frac{1}{2}. \quad \square$$

The constant would be sharp along a hypothetical infinite family in which the single-prime constituent is 2 and all other primes tend to equality with it. This is a prime-pattern condition, not an unconditional sharpness theorem. The inequalities were numerically checked for  $\omega = 3, \dots, 7$ .

## General pattern

The classification for  $\omega \in \{3, 4, 5\}$  exhibits a unified rule: the non-degenerate ratio  $\|\psi_{nd}\|_\infty / R^{1/(\omega-1)}$  is *bounded* if and only if there exist two primes from opposite sign-sides (from  $P_a \cup P_b$  vs.  $P_c$ , or from  $P_a$  vs.  $P_b$ ) that can be paired with a small  $\pm 1$   $\varphi$ -entry while the remaining large primes carry zero weight. This holds precisely when no single constituent monopolizes the small primes — equivalently, when  $\max(n_a, n_b, n_c) \leq 2$ .

## 6 Formal verification status

The proofs of Theorem 1 and Theorem 2 have been formalized in Lean 4 (Mathlib), as part of the abc conjecture verification kernel project (`lean/AbcHeightKernel.lean`, sections E3–E4, F3). Key items:

- `finite_prime_recip_sq_lt_one`: for any finite set  $P$  of primes,  $\sum_{p \in P} p^{-2} < 1$ . Proved by the telescoping argument above (zero `sorry`).
- `pasten_coeff_sq_sum`: the coefficient identity  $\sum_p (R/p)^2 = R^2 \sum_p p^{-2}$ .
- `pasten_det_lt_rad`:  $\sqrt{\sum_p (R/p)^2} < R$ .
- `minkowski_vaaler_pasten`: admitted as an axiom citing Vaaler [2]. Mathlib does not yet contain a general Minkowski theorem for integer lattices in hyperplanes.
- `pasten_prime_triple_arith` (F3):  $p \cdot q < 4r$  for prime triple  $p + q = r$ ,  $p = 2$ . Proved by `omega` after substitution.
- `pasten_rad_sqrt_lt_twice_r` (F3):  $\sqrt{pqr} < 2r$  for prime triples. Proved from the above via `nlinarith` and `Real.sqrt_lt_sqrt`. Zero `sorry`.

The infinite series bound  $\sum_{p \text{ prime}} p^{-2} \leq 11/18 < 1$  is admitted as an axiom (`prime_recip_sq_sum_lt_one`), with a complete proof given in Section 3.2 of the accompanying outsource document OB-09.

The non-degeneracy results (F3–F10) are currently in the discovery tier (`discovery/m2_directions/`); F5 (lattice membership) and F8 (universal divisibility) have been formalized in Lean 4 (`lean/AbcHeightKernel.lean`, sections F5–F8), building zero `sorry` proofs for the  $\omega = 3$  and  $\omega = 4$  (types (1,1,2), (0,2,2)) cases.

The upper-bound core for type (1, 1, 1) (Theorem 39) has been formalized with two Lean 4 theorems (zero `sorry`):

- `pasten_F16_111_key`:  $2q < p(p + q)$  for  $p \geq 2$ ,  $q > 0$  (proved by `nlinarith`).
- `pasten_F16_111_ratio_sq_lt_half`:  $q/(p(p + q)) < 1/2$  in  $\mathbb{R}$ .

The upper-bound core for type (1, 2, 1) (F12) has been formalized:

- `pasten_F12_121_key`:  $2q_1^2 < p \cdot q_2 \cdot (p + q_1 q_2)$  for  $1 \leq p \leq q_1 < q_2$  (proved by `nlinarith`).
- `pasten_F12_121_ratio_cube_lt_half`:  $q_1^2/(pq_2(p + q_1 q_2)) < 1/2$  in  $\mathbb{R}$ .

The upper-bound core for the c-even subfamily of type (1, 1, 2) (F14) has been formalized:

- `pasten_F14_112_ceven_key`:  $2p^2 < q(p + q)$  for  $1 \leq p < q$  (proved by `nlinarith`).
- `pasten_F14_112_ceven_ratio_cube_lt_half`:  $p^2/(q(p + q)) < 1/2$  in  $\mathbb{R}$ .

These three families cover the  $\omega \leq 4$  balanced-type upper-bound core.

For  $\omega = 5$ , the type (1, 2, 2) key inequality is also formalized:

- `pasten_F27_122_rho4_key`: for  $p+1 \leq q, r$  and  $r+1 \leq s$ ,  $p^3 < qrs$  (Nat, `nlinarith`).
- `pasten_F27_122_rho4_lt_half_real`:  $p^3/(2qrs) < 1/2$  in  $\mathbb{R}$  (Theorem 8 key inequality, Lean zero `sorry`).

For the universal pattern (Theorem 39), the general key lemma and the  $\omega = 6$  instance are formalized:

- `pasten_F28_key_gen`: for  $n \geq 1$  naturals each  $> p$ ,  $p^n < \prod x_i$  (Lean, `Finset.prod_le_prod + Nat.pow_lt_pow_left`).
- `pasten_F28_omega6_rho_lt_half`:  $p^4/(2qrst) < 1/2$  in  $\mathbb{R}$  ( $\omega = 6$  instance of the universal pattern, zero `sorry`).

The general bound  $\rho^4 < 1/6$  for all type  $(2, 2, 1)$  triples (F29) is formalized:

- `pasten_F29_221_key`:  $6q_1^3 < p_1p_2q_2r$  for  $p_1 \geq 2$ ,  $p_2 \geq p_1+1$ ,  $q_2 \geq q_1+1$ ,  $r \geq q_1q_2$  (`Nat, calc+nlinarith`).
- `pasten_F29_221_rho4_lt_sixth`:  $q_1^3/(p_1p_2q_2r) < 1/6$  in  $\mathbb{R}$  (`div_lt_one` pattern, zero `sorry`).

All proofs are zero `sorry`.

Additional lemmas added in the E10/E11/ $E_n$ /F31 extension:

- `pasten_E10_lb_key`: if  $\psi_q = \psi_r = 0$  and the  $\omega = 3$  lattice constraint holds, then  $W = 0$  (zero `sorry`).
- `pasten_E11_lb_key`: same for  $\omega = 4$  type  $(1, 1, 2)$  (zero `sorry`).
- `pasten_F31a`:  $(a+2)^a < (2a)^{a+4}$  for all  $a \geq 1$ ; algebraic core of the Quality- $\rho$  Joint Bound (zero `sorry`).
- `pasten_quality_lt_half_core`:  $a+b < ab$  for  $a \geq 2$ ,  $b \geq 3$ ; integer core of quality  $< 1/2$  (zero `sorry`).
- `pasten_F32a_gap`:  $q+2 \leq k \cdot q$  for all  $k \geq 2$ ,  $q \geq 3$ ; no non-degenerate norm lies strictly in  $(q, q+2)$  (F32a core, zero `sorry`).
- `pasten_F32a_upper`:  $\max(\max(0, q), q+2) = q+2$ ; norm of the second-minimum vector  $\varphi = (0, 1, 1)$  (zero `sorry`).
- `pasten_F32b_ineq`:  $2q+4 \leq q^2$  for  $q \geq 5$ ; algebraic core blocking  $p$ -term dominance in the F32b spectrum proof (zero `sorry`).
- `pasten_F33_gap`:  $N \leq k \cdot r_1$  for  $k \geq 2$ ,  $N \leq 2r_1$ ; integer core of second-minimum gap for type  $(1, 1, 2)$  (F33, Theorem 34, zero `sorry`).
- `pasten_En_lb_general`: for any Finsets  $P_a, P_b, P_c$  of primes satisfying the lattice constraint  $\sum_{P_a} f + \sum_{P_b} f = \sum_{P_c} f$ , if  $f = 0$  on  $P_b$  and  $f = 0$  on  $P_c$ , then  $\sum_{P_b} f = \sum_{P_a} f$  (degenerate condition for all  $\omega$ ; Lean, zero `sorry`). This is the first fully general- $\omega$  Lean formalization in this project.
- Four OB-13B integer kernel lemmas (zero `sorry`): cross-group norm bound;  $\text{median}^{\omega-1} \leq R$  for  $\omega = 3, 4$ ; within-group  $W \neq 0$ ; all in `lean/AbcHeightKernel.lean`.

- `pasten_nd_lb_large_prime_zero`:  $p_2 \leq p, px < p_2 \Rightarrow x = 0$ ; integer core of LB-1 (zero sorry).
- `pasten_nd_lb_single_prime_int`:  $k > 0, kz = 0 \Rightarrow z = 0$ ; abstract form of Lemma 18 (zero sorry).
- `pasten_nd_lb_omega2_ge_q`:  $q \leq q(v/g)$  for  $g \mid v$ ; integer core of  $\text{nd} \geq p_2$  in Theorem 20 (zero sorry).
- Three type  $(k, 1, 1)$  integer kernel lemmas for Theorem 23 (zero sorry): `pasten_nd_k11_r_branch` ( $r \leq rx$  for  $x \geq 1$ ;  $\varphi_r \neq 0$  branch); `pasten_nd_k11_valuation_branch` ( $qk \leq qkx$ ;  $\varphi_r=0$  branch); `pasten_nd_k11_norm_equals_qk` ( $\max(px, qkx) = qkx$  for  $p < q$ ;  $q$ -component dominates).
- `pasten_nd_km1_gcd_div_one`:  $\gcd(k/g, m/g) = 1$  for  $g = \gcd(k, m)$ ; integer core of the  $\varphi_r=0$  solution-family uniqueness in Theorem 25 (Mathlib `Nat.Coprime.div_gcd_coprime`, zero sorry).
- `pasten_nd_kmn_val_witness_constraint`:  $k \cdot (m/g) = m \cdot (k/g)$  for  $g = \gcd(k, m)$ ; confirms the valuation-regime witness  $(-m/g, k/g, 0)$  satisfies the constraint for *any*  $n$  (Corollary 26), using `Nat.mul_div_assoc` twice. Zero sorry.
- `pasten_nd_kmn_val_n_zero`:  $n \cdot 0 = 0$  for all  $n$ ; captures the key  $n$ -independence: the  $r$ -coordinate of the valuation-regime witness is zero, so the  $c$ -exponent  $n$  does not appear (Corollary 26). Zero sorry.

## 7 Conclusion

We have proved that for squarefree coprime triples  $(a, b, c)$  with  $a + b = c$  and  $\omega(abc) \geq 2$ , the Pasten lattice satisfies  $\det(F(a, b)) < \text{rad}(abc)$ , and hence Vaaler's subspace lattice theorem [2] gives a nonzero vector with  $\|\psi\|_\infty \leq \text{rad}(abc)^{1/(\omega-1)}$ .

Beyond the determinant bound, we have established a structural classification of non-degeneracy for all partition types with  $\omega \leq 5$ , proved that all types with  $\omega \geq 6$  satisfy the criterion of Theorem 37 whenever the required prime-pattern family exists, and proved unconditional upper bounds for every universally-bounded type (Theorems 38 and 39). The central structural results are: (a) the Universal Divisibility Lemma ( $q \mid \psi_q$  for all  $q \in P$ , proved by elementary divisibility from the lattice constraint); (b) the  $\varphi$ -coordinate change reducing  $F(a, b)$  to the simplest integer hyperplane lattice; (c) the optimal crossing formula (Theorem F10): the minimum non-degenerate norm is the second smallest of  $\{\min P_a, \min P_b, \min P_c\}$ , achieved by any cross-group prime pair; (d) the universal lower bound (Theorem E<sub>n</sub>): every non-degenerate  $\psi$  satisfies  $\|\psi\|_\infty \geq \text{nd}$ , proved by a clean F8+constraint+Wronskian argument and Lean-formalized for  $\omega = 3, 4$  (E10/E11), numerically verified for all types  $\omega \leq 7$ ; (e) a complete table of non-degeneracy upper bounds for all partition types  $\omega \in \{3, 4, 5\}$  (Table in Theorem 38); (f) the unifying upper bound  $\rho^{\omega-1} < 1/2$  for *all* balanced types and  $\omega \geq 3$  (Theorem 39, F28), with the general key lemma Lean-formalized; and (g) the general bound  $\rho^4 < 1/6$  for *all* type  $(2, 2, 1)$  triples (F29, Lean-formalized), with conditional

sharpness at  $6^{-1/4}$ ; and (h) the corrected bound  $\rho^4 < 1/2$  for *all* type  $(2, 1, 2)$  triples (F30, Lean-formalized, correcting F25), with conditional sharpness equal to the  $(1, 2, 2)$  constant by  $a \leftrightarrow b$  symmetry; and (i) the Quality- $\rho$  Joint Bound (F31, Theorem 5): for every non-trivial squarefree triple of any bounded- $\rho$  type,  $\text{quality} + \rho^{\omega-1} < 1$ , proved by combining  $\text{quality} < 1/2$  (Proposition 4) with  $\rho^{\omega-1} < 1/2$  (Theorems F12–F30); and (j) the second-minimum and complete spectrum results for type  $(1, 1, 1)$ : the second minimum non-degenerate norm equals  $r$  (F32a, Theorem 31), and for every twin-prime triple with  $q \geq 5$  the full spectrum of achievable non-degenerate norms is exactly  $\{kq : k \geq 1\} \cup \{kr : k \geq 1\}$  (F32b, Theorem 32), both Lean-formalized; this union of two rays has no Frobenius number (Corollary 33); and second minimum  $\min(r_2, q, 2r_1)$  for type  $(1, 1, 2)$  (F33, Theorem 34), Lean-formalized; and (k) the non-squarefree upper bound (OB-13B, Theorem 15):  $\text{nd}(a, b) \leq v_{\max} \cdot R^{1/(\omega^*-1)}$  for all coprime triples, proved via the cross-group vector construction, with a tighter median form (Corollary 16), and the within-group construction showing that the squarefree  $E_n$  lower bound fails for non-squarefree triples (counterexample (72, 11, 83)); (l) the universal lower bound (Theorem 19):  $\text{nd}(a, b) \geq p_2$  (second-smallest prime in  $\text{rad}(abc)$ ) for all coprime triples, squarefree or not, proved by a single-prime-constraint argument (Lemma 18), tight at  $(1, 2, 3)$ ; and (m) the exact formula for  $\omega^* = 2$  triples (Theorem 20, Corollary 21):  $\text{nd}(a, b) = \max(p \cdot w/g, q \cdot v/g)$  where  $g = \gcd(v, w)$ , verified on all seven  $\omega^* = 2$  examples up to  $c \leq 1000$ , Lean-formalized; and (ag) the complete norm spectrum for  $\omega^* = 2$  triples (Corollary 22): the achievable norms of non-degenerate vectors are exactly  $\{k \cdot \text{nd}(a, b) : k \in \mathbb{Z}_{>0}\}$ , so all successive minima satisfy  $\lambda_k = k \cdot \text{nd}$ —the lattice is effectively one-dimensional and its spectrum is a pure arithmetic progression; and (n) the exact formula for type  $(k, 1, 1)$  non-squarefree triples (Theorem 23):  $\text{nd}(a, b) = \min(r, qk)$  for  $P_a = \{p^k\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r\}$  with  $p < q < r$ , with two regimes (valuation-dominant  $qk < r$  and pairwise-dominant  $qk \geq r$ ) and confirming  $\text{nd} \geq p_2 = q$  (Theorem 19); and (ah) the second successive minimum for type  $(k, 1, 1)$  triples (Theorem 24): in the valuation regime  $\lambda_2 = \min(r, 2qk)$ ; in the pairwise regime  $\lambda_2 = \min\{N \in \{2r, qk, q(k-1)\} : N > r\}$  (three explicit achievability vectors, gap proved by a parametric line argument); verified on 55 triples, zero failures (T87); verified numerically (T68) and Lean-formalized (three arithmetic kernel lemmas, zero `sorry`); and (ai) the second successive minimum for type  $(k, m, 1)$  triples in the valuation regime (Theorem 27):  $\lambda_2 = \min(r, 2N_0)$  where  $N_0 = \max(pm/g, qk/g)$ ,  $g = \gcd(k, m)$ , proved by the same three-way split as Theorem 24 (the  $\varphi_r = 0$  branch generates norms  $\{tN_0 : t \geq 1\}$  by primitivity of its generator, any  $|\varphi_r| \geq 1$  vector gives norm  $\geq r \geq \min(r, 2N_0)$ ); verified on 40 triples ( $p \leq 7$ ,  $k, m \in \{2, \dots, 9\}$ , zero failures, T88b); and (o) the exact formula for general type  $(k, m, 1)$  non-squarefree triples (Theorem 25):  $\text{nd}(a, b) = \min(r, \max(pm/g, qk/g))$  where  $g = \gcd(k, m)$ , subsumes Theorems 23 ( $m = 1$ ) and the squarefree  $E_n$  value ( $k = m = 1$ ), with equal-valuation regime ( $k = m$ ) always realising  $\text{nd} = p_2$ ; verified numerically on 14 named examples (T69); and (p) a universality corollary (Corollary 29): for one-prime-per- $a, b$ -group triples with  $r_1 = \min P_c > q > p$ , the formula  $\text{nd} = \min(r_1, \max(pm/g, qk/g))$  holds regardless of how many primes  $c$  has or their valuations — only the smallest  $c$ -prime matters; and (q) a valuation-regime independence result (Corollary 26): when  $\max(pm/g, qk/g) \leq r$ , the formula  $\text{nd}(a, b) = \max(pm/g, qk/g)$  holds for *all*  $n = v_r(c)$ , not just  $n = 1$  — the witness  $(-m/g, k/g, 0)$  satisfies the constraint for any  $n$ , so the  $c$ -exponent is entirely invisible in the valuation regime (verified numerically on 52 cases with  $n \geq 2$ , T71); and (r) a characterisation of the pure-formula boundary (Re-



mark 10): when the group- $c$  prime is not the largest in the triple, the three-pure formula can fail and a fully mixed witness is required, exemplified by (5, 27, 32) where  $\text{nd} = 6$  beats all pure-branch minima ( $N_0 = N_1 = 15$ ,  $N_2 = 25$ ); and (s) a complete pairwise-regime characterisation (Theorem 28): for one-prime-per-group triples in the pairwise regime  $N_{p_L=0} > p_L$ , the exact formula is  $\text{nd}(a, b) = \min(N, \max(p_L, B))$  where  $B$  is the minimum weighted norm over  $|\varphi_{p_L}| = 1$  solutions — simultaneously giving  $\text{nd} = p_L$  in the typical case ( $B \leq p_L$ , verified 223/229 cases, T72) and recovering the precise value in the 6 exceptional cases (e.g. (169, 343) with  $\text{nd} = N = 21$  due to a gcd obstruction); and (t) a systematic scan (Remark 12, T73) of all quality- $> 1$  one-prime-per-group triples with  $a + b \leq 5 \times 10^5$ , yielding exactly 10 such triples — all correctly classified by the exact formula, with four exhibiting  $\text{nd} < p_L$  via the  $\text{val}(\varphi_{p_L} = 0)$  regime, and confirming that  $\text{nd}$  and quality are structurally independent ( $\text{nd}$  depends on  $(p_a, p_b, p_c, k, m)$  while quality depends on  $n$ ); Items (u)–(ae) below concern the valuation-coordinate lattice  $F_{\text{val}}$  and are candidate formulas with finite all-nonzero-case verification; they are not unconditional theorems and do not determine the original Pasten lattice minimum in general. and (u) the exact  $\text{nd}$  formula for  $\omega^* = 4$  triples with two  $c$ -group primes (Candidate Formula 1): for  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r, s\}$ ,  $\text{nd}(a, b) = \min(\text{nd}_3(p^k, q^m, r^j), \text{nd}_3(p^k, q^m, s^n), \max(q, B_m), \max(p, B_k))$  where  $B_x$  is the two-dimensional Bezout minimum for  $j\varphi_r + n\varphi_s = x$ ; the  $\varphi_q = 0$  branch wins in 25 out of 431 tested cases (T74), confirming that the four-branch formula subsumes the  $\omega^* = 3$  characterization; and (v) the exact  $\text{nd}$  formula for type (2, 1, 1) triples (Candidate Formula 2): for  $P_a = \{p, q\}$ ,  $P_b = \{r\}$ ,  $P_c = \{s\}$ , the minimum is taken over four branches corresponding to zeroing one prime-group slot —  $N_{s=0}$  ( $\text{Pa} \times \text{Pb}$  interaction, subsumes within-group and pairwise norms),  $N_{r=0}$  ( $\text{Pa} \times \text{Pc}$  interaction), and two  $\omega^* = 3$  sub-problems  $\text{nd}_3(p^{k_1}, r^m, s^n)$ ,  $\text{nd}_3(q^{k_2}, r^m, s^n)$ ; verified on 672 triples (T77, 0 failures), with  $N_{s=0}$  dominant in 87% of cases; and (w) the exact  $\text{nd}$  formula for type (1, 2, 1) triples (Candidate Formula 3): for  $P_a = \{p\}$ ,  $P_b = \{q, r\}$ ,  $P_c = \{s\}$ , the same four-branch structure with  $P_a \leftrightarrow P_b$  exchanged —  $N_{s=0}$  ( $\text{Pb} \times \text{Pa}$  interaction),  $N_{p=0}$  ( $\text{Pb} \times \text{Pc}$  interaction), and two  $\omega^* = 3$  sub-problems; verified on 1 043 triples (T78\_formula, 0 failures), with  $N_{s=0}$  dominant in 90% of cases; and (x) the exact  $\text{nd}$  formula for type (2, 2, 1) triples (Candidate Formula 4): for  $P_a = \{p, q\}$ ,  $P_b = \{r, s\}$ ,  $P_c = \{u\}$ , the minimum is taken over five branches —  $N_{u=0}$  (the  $P_a$ – $P_b$  homogeneous interaction, zeroing  $\varphi_u$ ), two type-(1, 2, 1) sub-problems (zeroing  $\varphi_p$  or  $\varphi_q$  via Candidate Formula 3), and two type-(2, 1, 1) sub-problems (zeroing  $\varphi_r$  or  $\varphi_s$  via Candidate Formula 2); verified on 1 425 triples with  $a, b \leq 300$  (T79, spot-check 60, 0 failures), with  $N_{u=0}$  dominant in 98.6% of cases; and (y) the exact  $\text{nd}$  formula for type (1, 2, 2) triples (Candidate Formula 5): for  $P_a = \{p\}$ ,  $P_b = \{q, r\}$ ,  $P_c = \{s, t\}$ , the minimum over five branches —  $N_{p=0}$  ( $P_b$ – $P_c$  homogeneous interaction, zeroing  $\varphi_p$ ), two type-(1, 1, 2) sub-problems (zeroing  $\varphi_q$  or  $\varphi_r$  via Candidate Formula 1), and two type-(1, 2, 1) sub-problems (zeroing  $\varphi_s$  or  $\varphi_t$  via Candidate Formula 3); verified on 996 triples with  $a, b \leq 150$  (T80, spot-check 50, 0 failures), with  $N_{p=0}$  dominant in 87.2% of cases; and (z) the exact  $\text{nd}$  formula for type (2, 1, 2) triples (Candidate Formula 6), obtained from Candidate Formula 5 by the  $P_a \leftrightarrow P_b$  label swap:  $N_{r=0}$  ( $P_a$ – $P_c$  interaction) plus two type-(1, 1, 2) and two type-(2, 1, 1) sub-problems; and (aa) the exact  $\text{nd}$  formula for type (1, 1, 3) triples (Candidate Formula 7): for  $P_a = \{p\}$ ,  $P_b = \{q\}$ ,  $P_c = \{r, s, t\}$ , the minimum of five branches — three type-(1, 1, 2) sub-problems (zeroing each  $c$ -prime in turn via Candidate Formula 1) and two one-sided norms  $N_{p=0}$ ,  $N_{q=0}$ ; verified on 729 triples with  $a, b \leq 300$  (T81, spot-check 60, 0 failures), with the  $\varphi_t = 0$

branch dominant (91.8%); and (ab) the exact nd formula for type (1, 3, 1) triples (Candidate Formula 8): for  $P_a = \{p\}$ ,  $P_b = \{q, r, s\}$ ,  $P_c = \{t\}$ , the minimum of five branches — a  $P_a \times P_b$  homogeneous norm  $N_{t=0}$  (zeroing  $\varphi_t$ ,  $W \neq 0$ ), a  $P_b \times P_c$  interaction norm  $N_{p=0}$  (zeroing  $\varphi_p$ ,  $\varphi_q + \varphi_r + \varphi_s \neq 0$ ), and three type-(1, 2, 1) sub-problems (zeroing each  $b$ -prime via Candidate Formula 3); verified on 843 triples with  $a, b \leq 300$  (T82, spot-check 60, 0 failures), with  $N_{t=0}$  dominant (95.3%); and (ac) the exact nd formula for type (3, 1, 1) triples (Candidate Formula 9), obtained from Candidate Formula 8 by the  $P_a \leftrightarrow P_b$  label swap; the mirror case was also checked directly on all 843 type-(3, 1, 1) triples with  $a, b \leq 300$  (T82-type311, 0 failures); and (ad) the *zero-coordinate minimizer property* (Candidate Formula 10): for  $\omega^* \geq 4$ , the minimum non-degenerate valuation-lattice norm is achieved by a vector with at least one prime coordinate equal to zero in all surveyed cases, so  $\text{nd}_{\text{val}}(a, b) = \min_p B_p$  on those samples (minimum over all zero- $\varphi_p$  sub-lattices); the squarefree case is proved, while  $\omega^* = 4, 5, 6$  non-squarefree cases are finite-verification claims: non-squarefree  $\omega^* = 6$ : 1959 triples  $a, b \leq 200$ , all three partition classes  $\{4, 1, 1\}$ ,  $\{3, 2, 1\}$ ,  $\{2, 2, 2\}$ , 0 violations (T85bis); squarefree  $\omega^* = 6$ : proved by Theorem 30. This fails for non-squarefree  $\omega^* = 3$ : the triple (5, 27, 32) achieves  $\text{nd}_{\text{val}} = 6$  via an all-nonzero vector, while every zero-coordinate branch gives norm  $\geq 15$ ; and (ae) the *general recursive valuation-nd algorithm* (Candidate Formula 11): for any coprime triple with  $\omega^* \geq 3$  primes, the candidate formula is  $\text{nd}_{\text{val}}(a, b) = \min_p B_p$  (each  $B_p$  zeroing one prime coordinate, computed recursively from the one-prime-per-group base case); it is proved in the squarefree case, and verified on finite samples for non-squarefree  $\omega^* = 4, 5, 6$ , including  $\omega^* = 5$  (all six partition types) and  $\omega^* = 6$  (all three partition classes, non-squarefree 1959 triples  $a, b \leq 200$ , 0 violations, T85bis; squarefree case from Theorem 30); and (af) the *squarefree minimizer support theorem* (Theorem 30): for every squarefree coprime triple of any  $\omega^* \geq 3$ , every nd-minimizer satisfies  $\varphi_p = 0$  for all  $p > m_2$  (second group minimum), so at least  $\omega^* - 2$  coordinates are forced to zero by integrality; in particular the zero-coordinate property holds unconditionally for all squarefree triples; and (aj) the universal second-minimum formula for all squarefree triples with  $\omega^* \geq 3$  (Theorem 35):  $\lambda_2(a, b) = \min(2p_2, p_3)$  where  $p_1 < p_2 < p_3$  are the three smallest primes of  $\text{rad}(abc)$ , proved by an integrality gap argument reducing any near-minimizer to the two-prime sub-lattice; unifies Theorem 31 ( $\omega^* = 3$ , where  $p_3 < 2p_2$  always) and Theorem 34 (type (1, 1, 2), where the  $q$ -branch is dominated); verified on 778 squarefree  $\omega^* = 4$  triples, 42 at  $\omega^* = 5$ , 12 at  $\omega^* = 6$  (zero failures, T90b); and (ak) the third successive minimum formula for squarefree type-(1, 1, 2) triples (Theorem 36):  $\lambda_3(a, b) = \min(2p_2, p_4)$  if  $p_3 < 2p_2$ , and  $\lambda_3(a, b) = \min(3p_2, p_3)$  if  $p_3 \geq 2p_2$ , where  $p_4$  is the fourth-smallest prime of  $\text{rad}(abc)$ ; proved by extending the integrality gap argument to the next branch — the “losing” branch of the  $\lambda_2$  competition always yields  $\lambda_3$ ; verified on 360 squarefree type-(1, 1, 2) triples with  $c \leq 500$  (zero failures, T94b).

The non-degenerate ratio  $\|\psi_{\text{nd}}\|/R^{1/(\omega-1)}$  is bounded for exactly the “balanced” partition types (both other constituents non-empty with available small-prime crossings), and grows without bound for the “isolated” types where both non-dominant constituents are forced-large single primes. For  $\omega \geq 6$ , unboundedness follows from Theorem 37 once the required prime-pattern family exists. The unconditional upper bounds for the bounded types are collected in Theorem 38. Their sharpness depends on corresponding infinite prime-pattern families and remains conditional.

We also prove a quality boundary for the squarefree subfamily (Proposition 4): quality

$> 1/2$  if and only if  $a = 1$ . Within bounded partition types, quality and  $\rho$  are negatively correlated ( $r \approx -0.76$  for types  $(1, 1, 2)$  and  $(1, 2, 1)$ ): the triples closest to the extremal  $\rho$  bound have quality  $\rightarrow 1/3$ , while the highest-quality non-trivial squarefree triples ( $a = 2$ ,  $b \rightarrow \infty$ ) have  $\rho \rightarrow 0$ . This quality- $\rho$  trade-off is a provable structural feature, not a numerical artifact.

**Honest scope.** None of these results constitute progress toward the abc conjecture or SDC. The Minkowski bound is  $\eta_R = 1/(\omega - 1)$  relative to  $\text{rad}(abc)$ , whereas SDC requires  $\eta_c < 1$  relative to  $c$  for *all*  $\varepsilon > 0$  — an exponential improvement. The classification of degenerate vs. non-degenerate types is structural geometry of Pasten’s lattice; it does not produce bounds on  $c$ . The abc conjecture remains open.

This is a structural result about the geometry of Pasten’s lattice construction, not progress toward abc. The abc conjecture requires  $\|\psi\|_\infty = o(c^\varepsilon)$  for all  $\varepsilon > 0$  (Pasten’s SDC), and the gap between our bound and SDC is the full depth of the problem. We have made no use of any abc-equivalent hypothesis, known abc triples, or the Szpiro conjecture.

## References

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